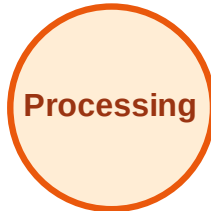


BFS Traversal

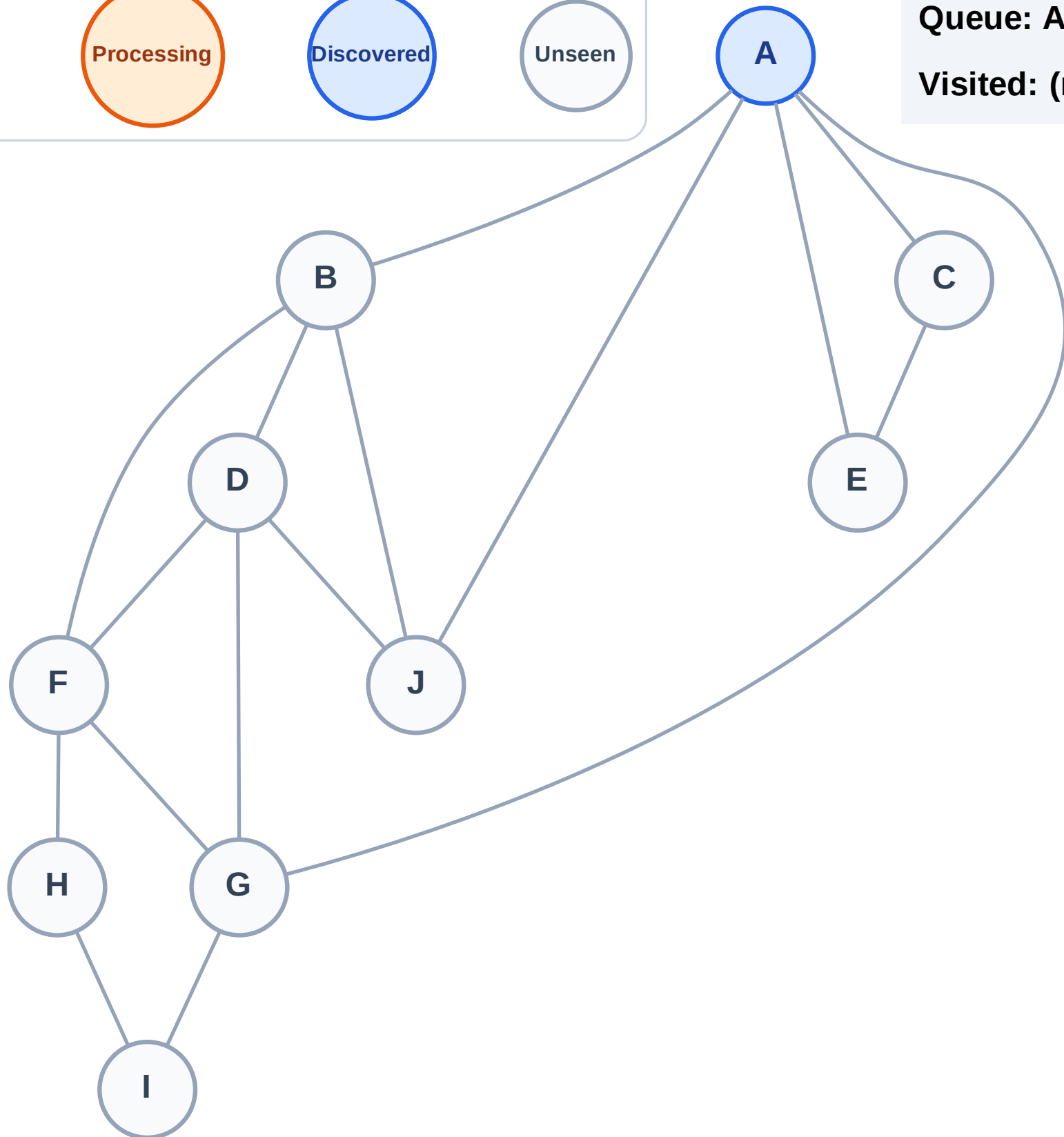
Step 1: Start at A

Legend



Queue: A

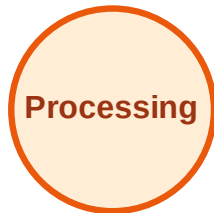
Visited: (none)



BFS Traversal

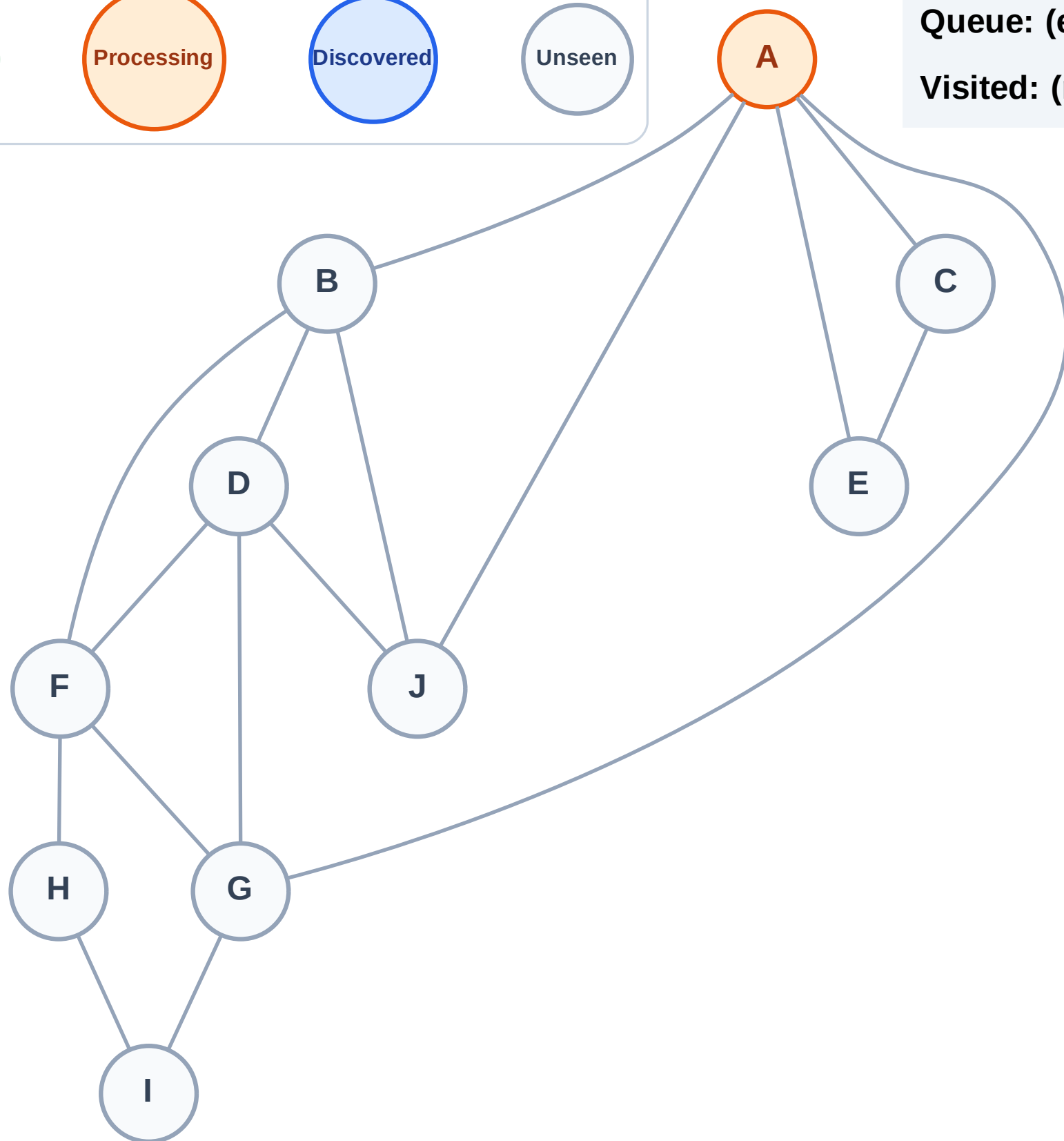
Step 2: Process node A

Legend



Queue: (empty)

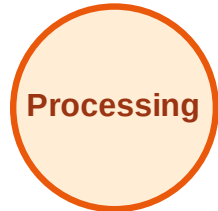
Visited: (none)



BFS Traversal

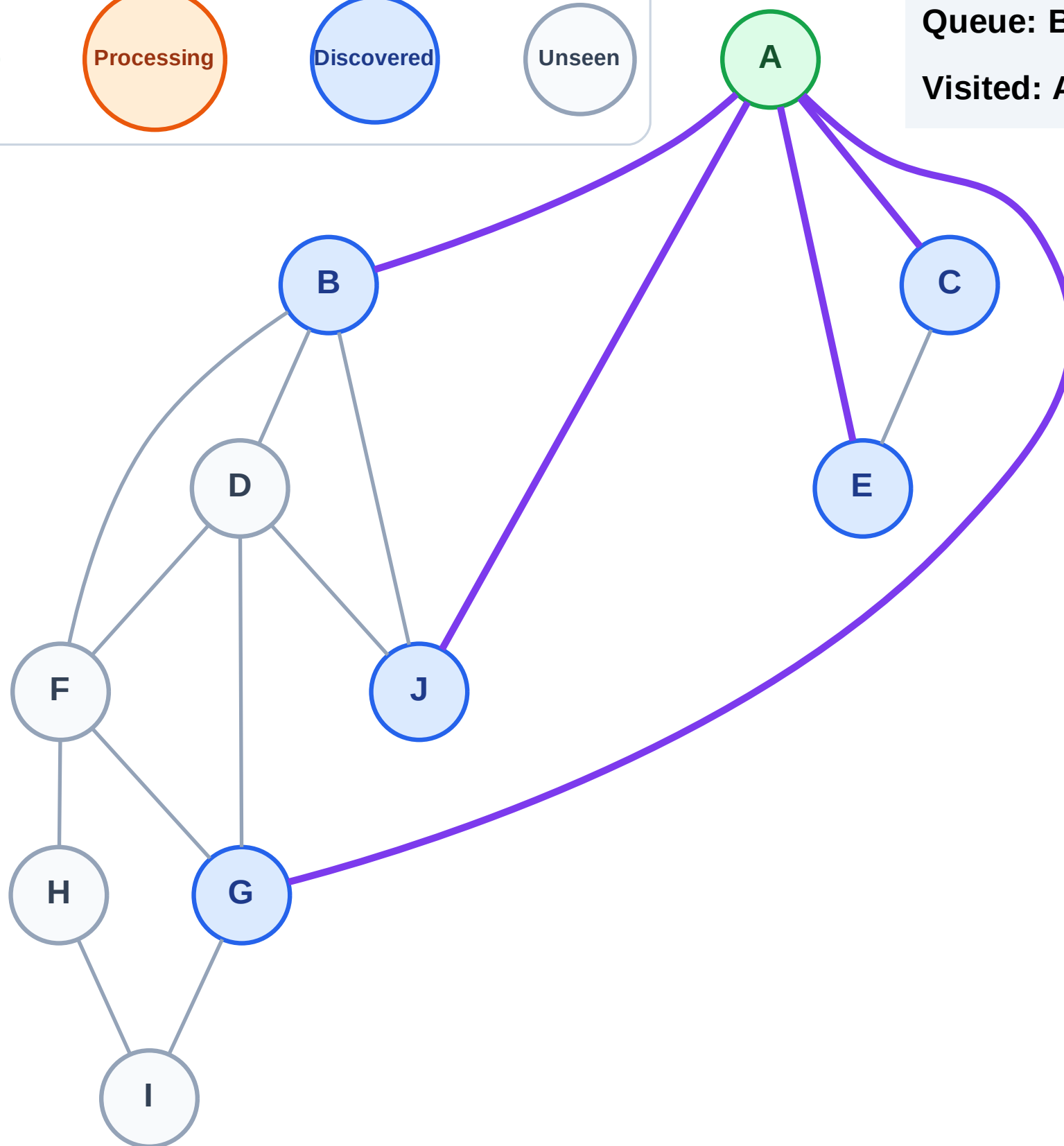
Step 3: Discover B, C, E, G, J from A

Legend



Queue: B → C → E → G → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

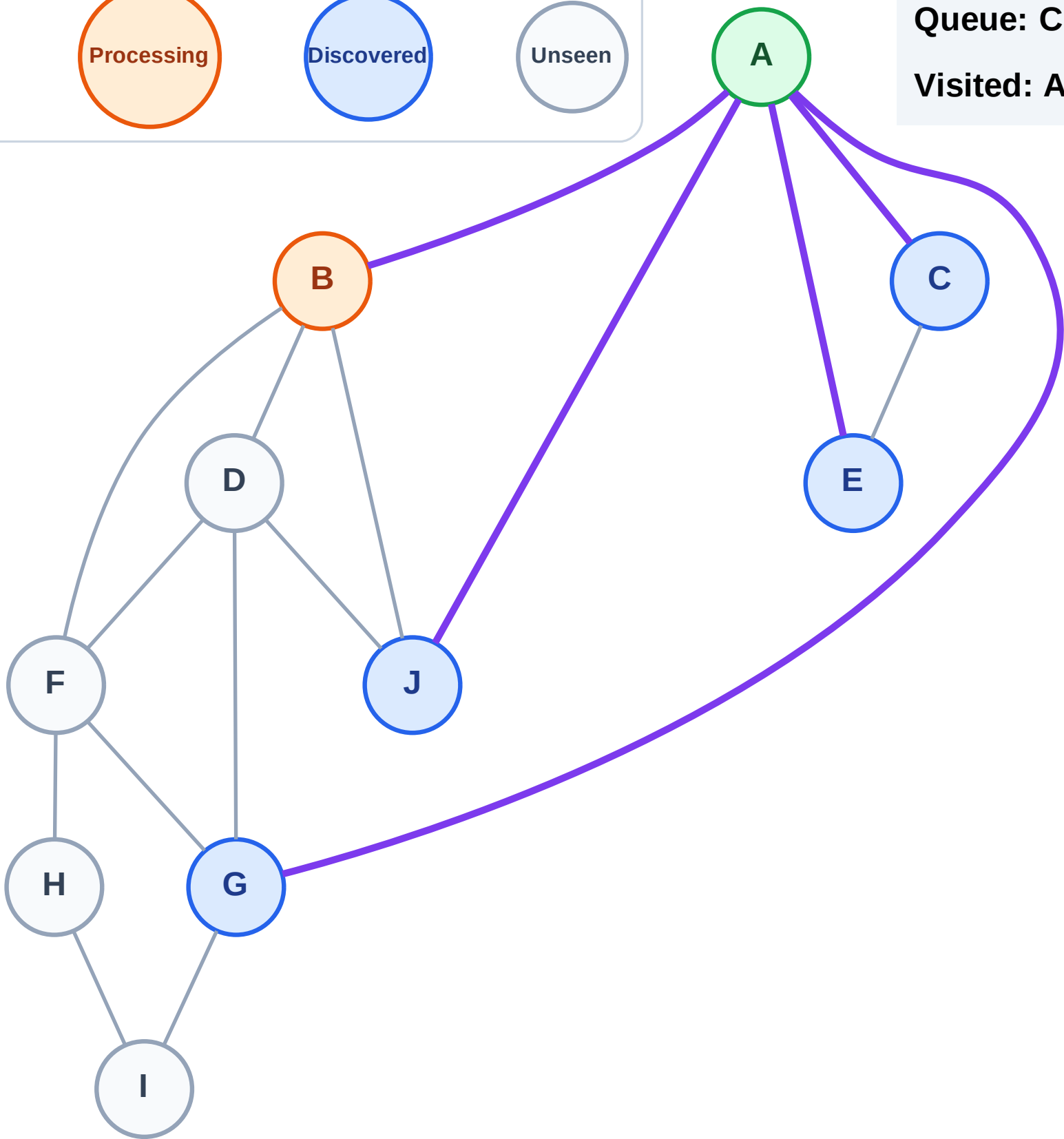
Processing

Discovered

Unseen

Queue: C → E → G → J

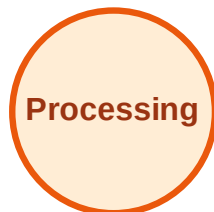
Visited: A



BFS Traversal

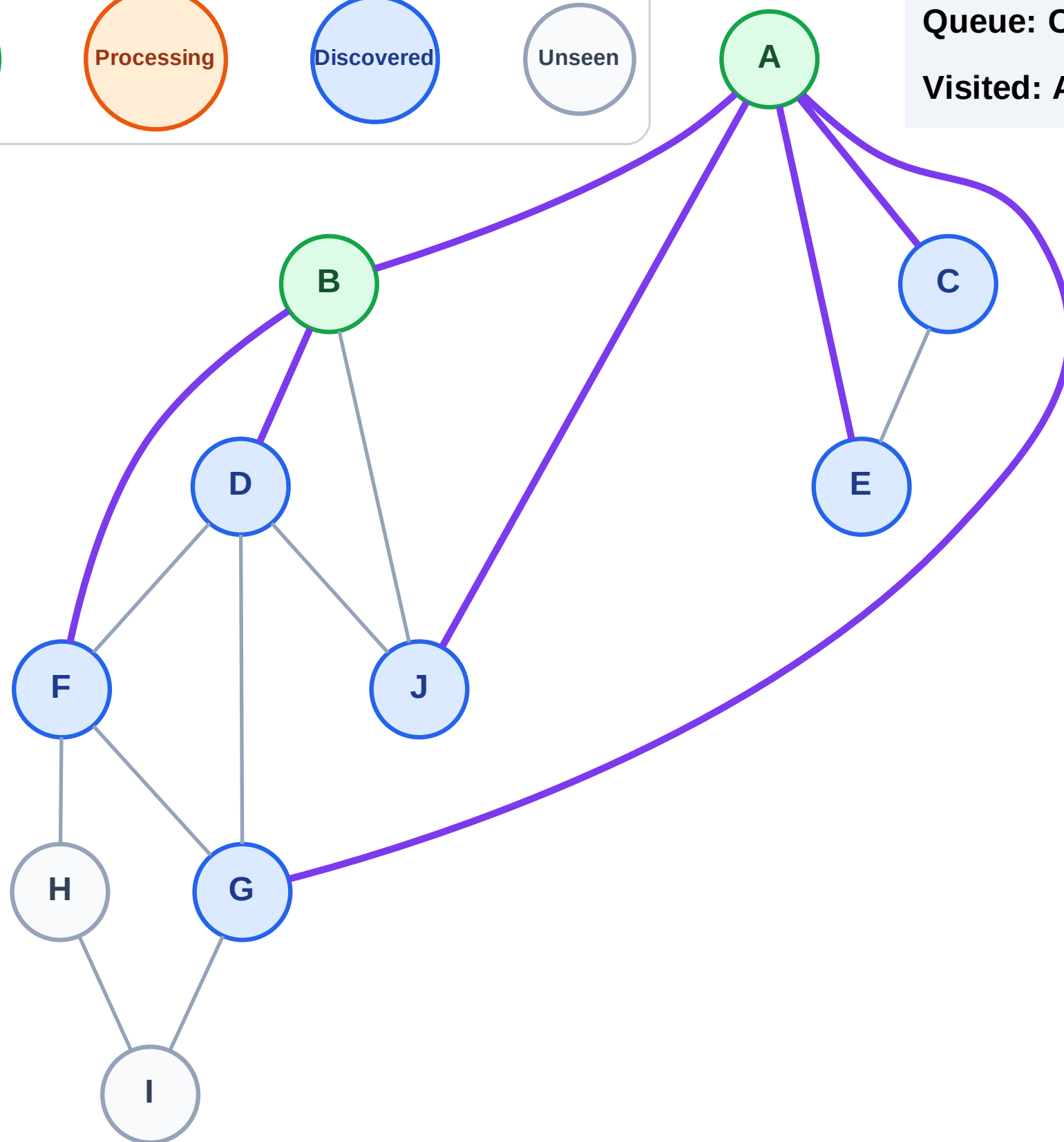
Step 5: Discover D, F from B

Legend



Queue: C → E → G → J → D → F

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

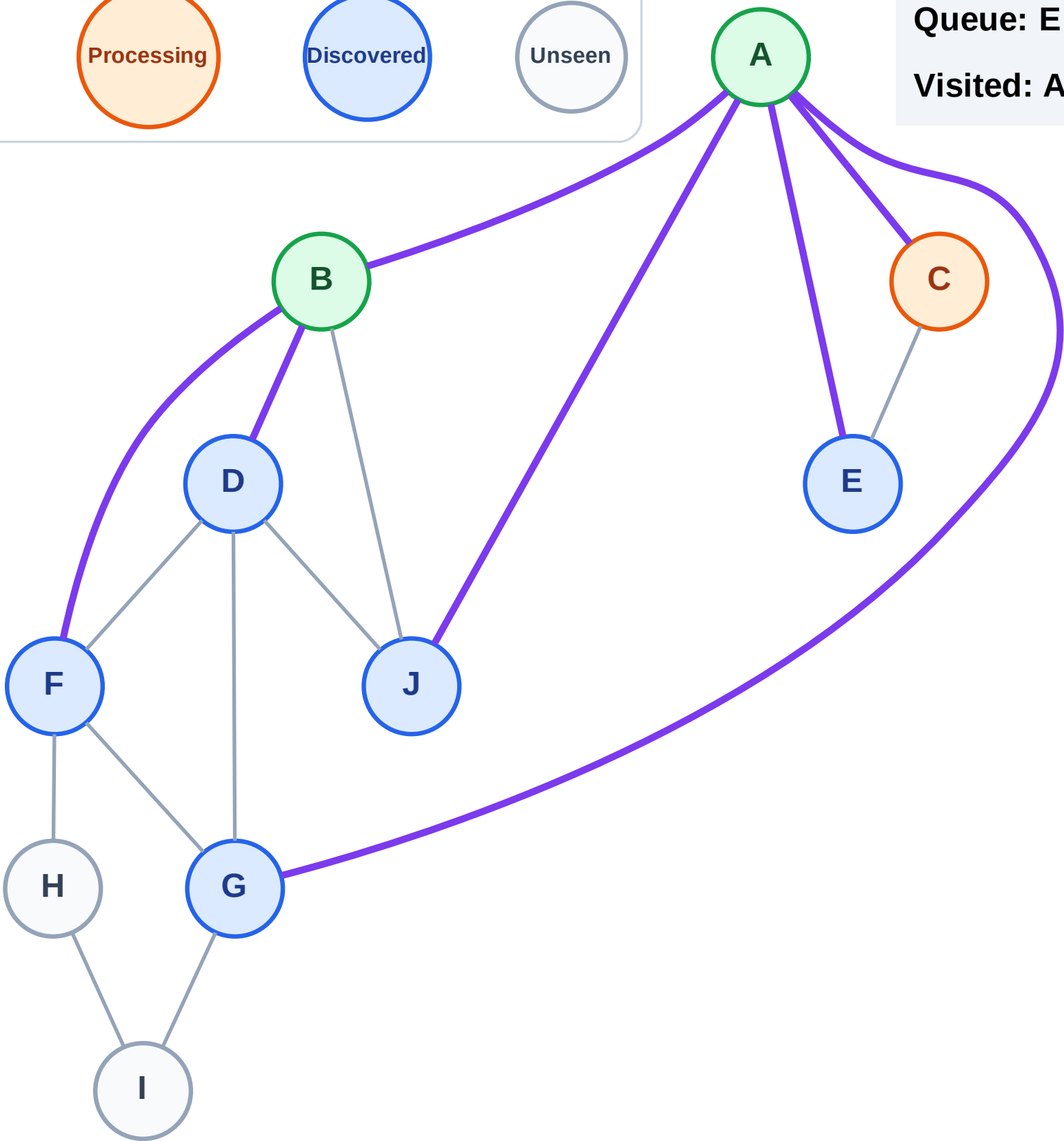
Processing

Discovered

Unseen

Queue: E → G → J → D → F

Visited: A, B



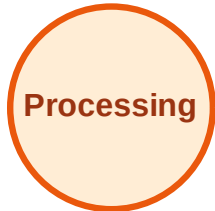
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



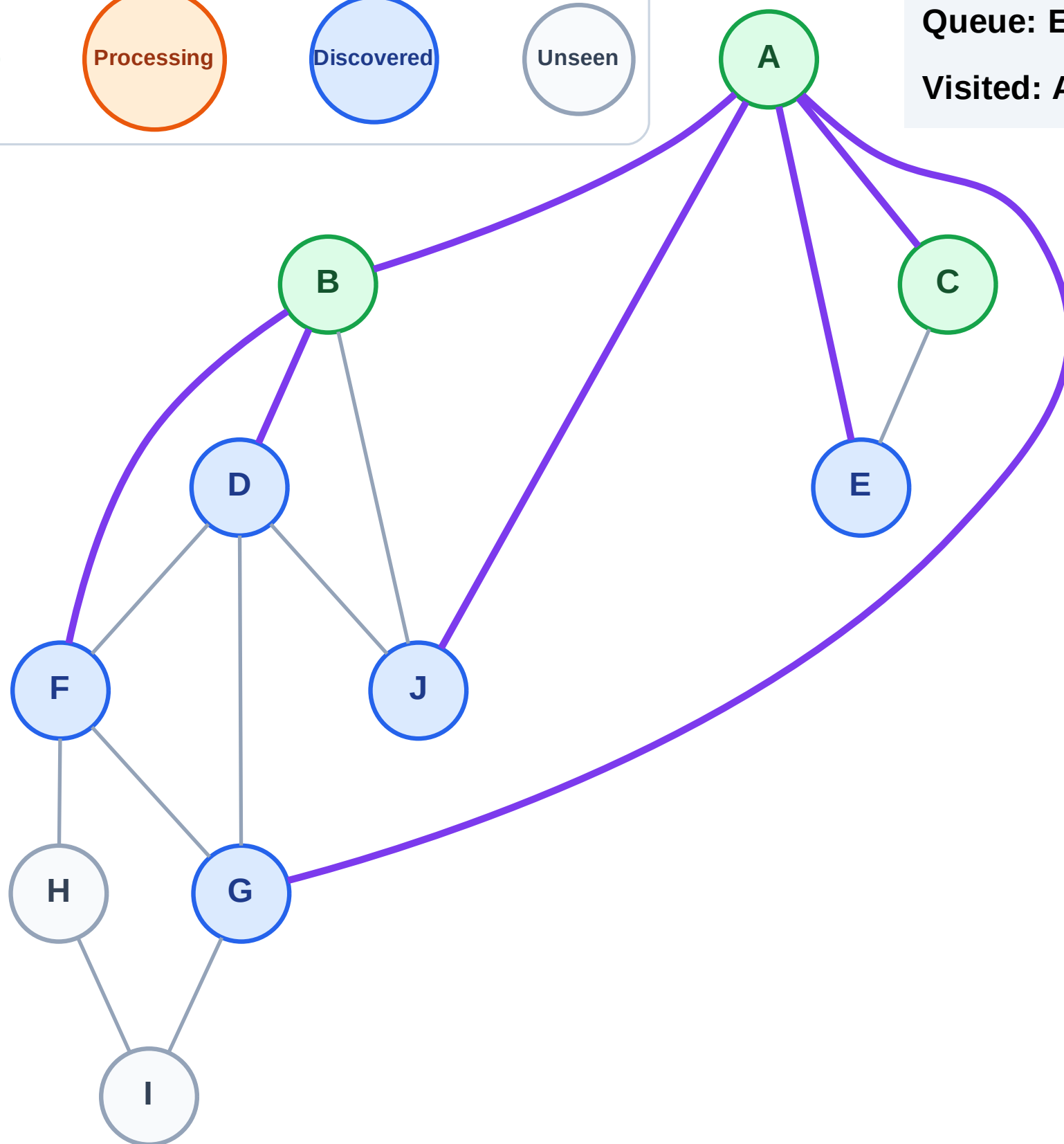
Discovered



Unseen

Queue: E → G → J → D → F

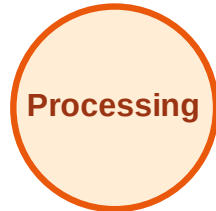
Visited: A, B, C



BFS Traversal

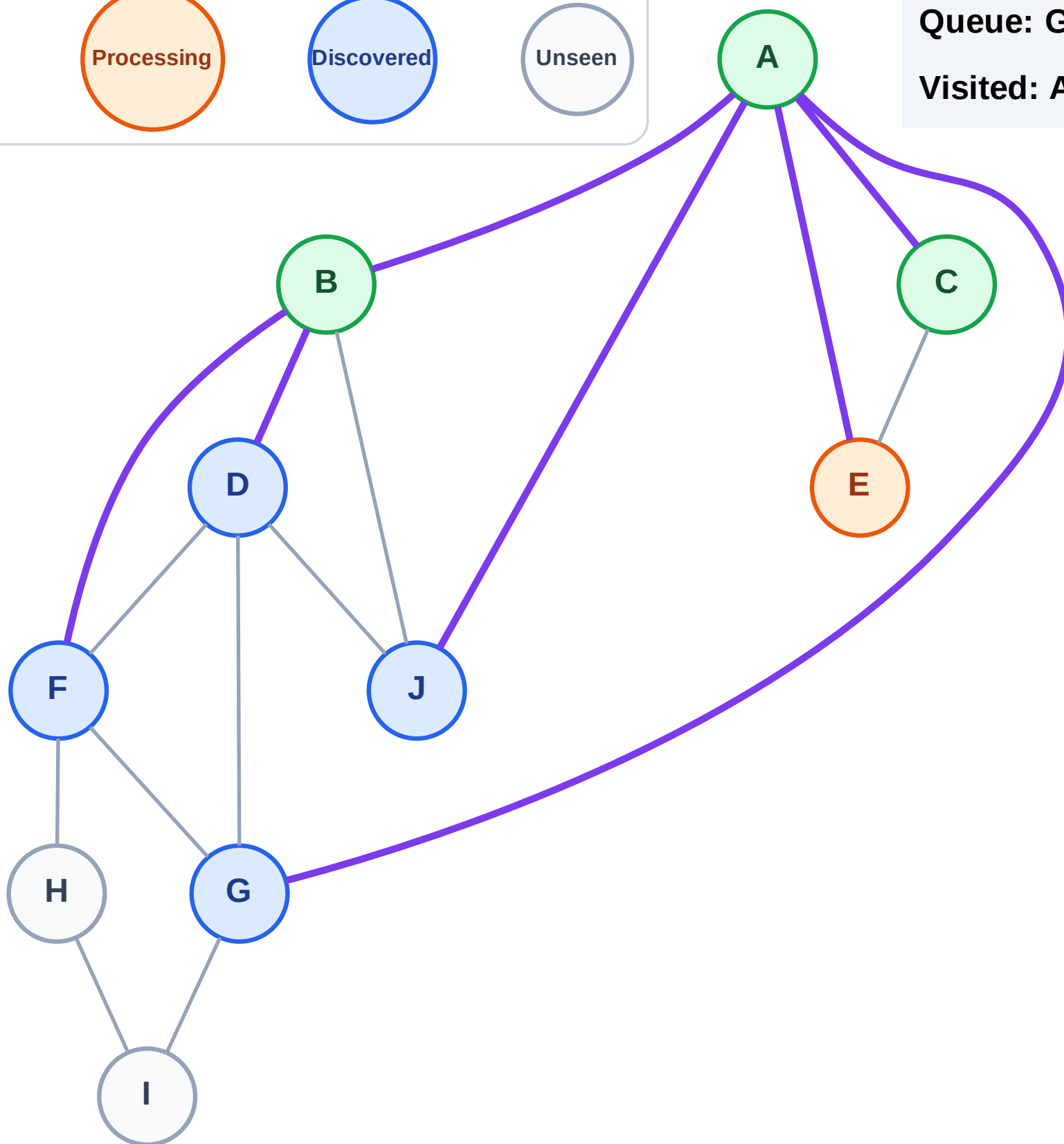
Step 8: Process node E

Legend



Queue: G → J → D → F

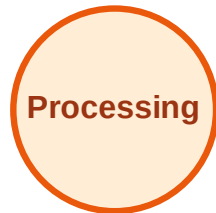
Visited: A, B, C



BFS Traversal

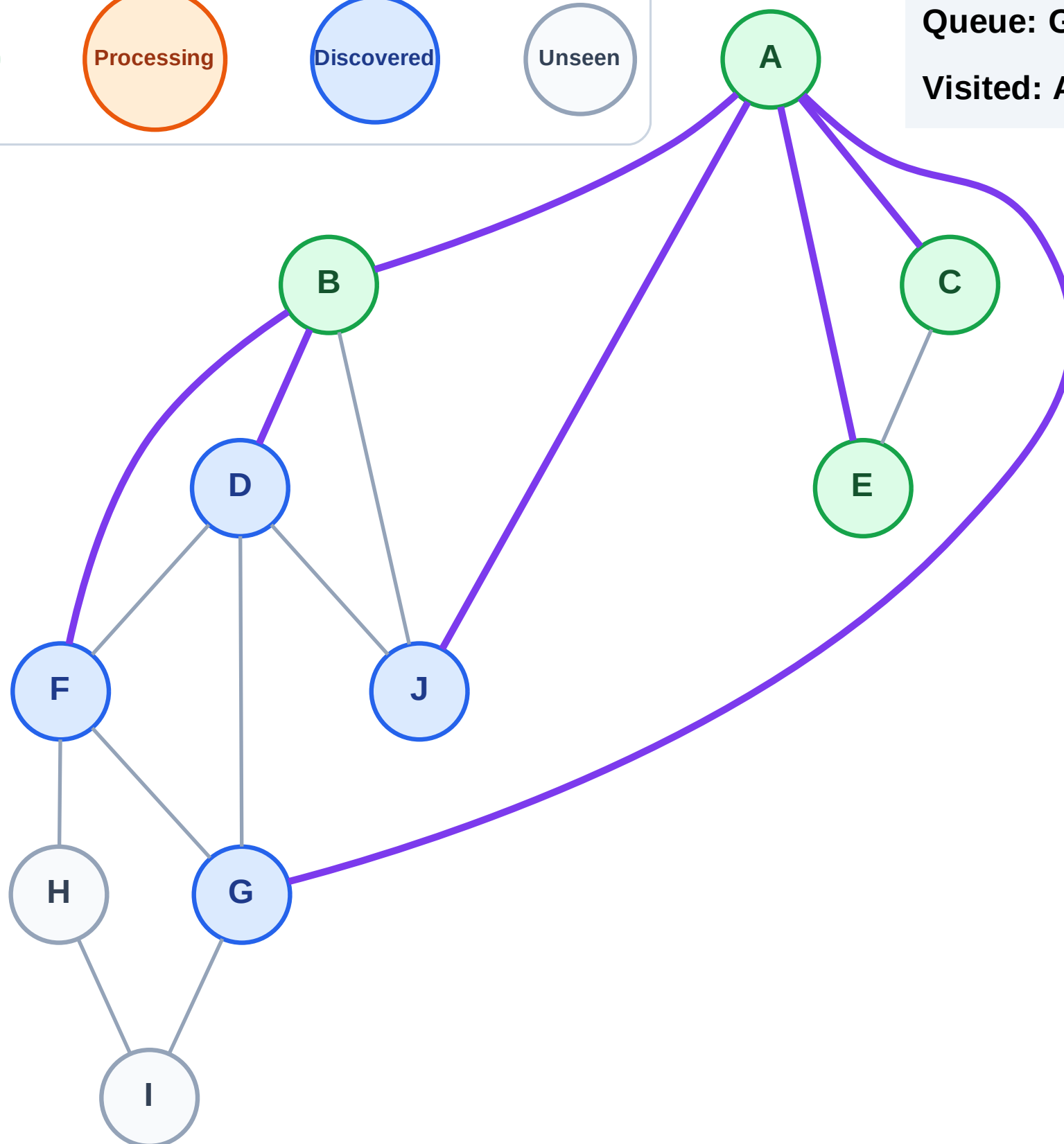
Step 9: No new neighbors from E

Legend



Queue: G → J → D → F

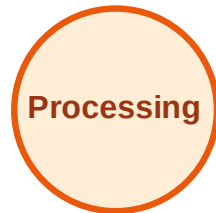
Visited: A, B, C, E



BFS Traversal

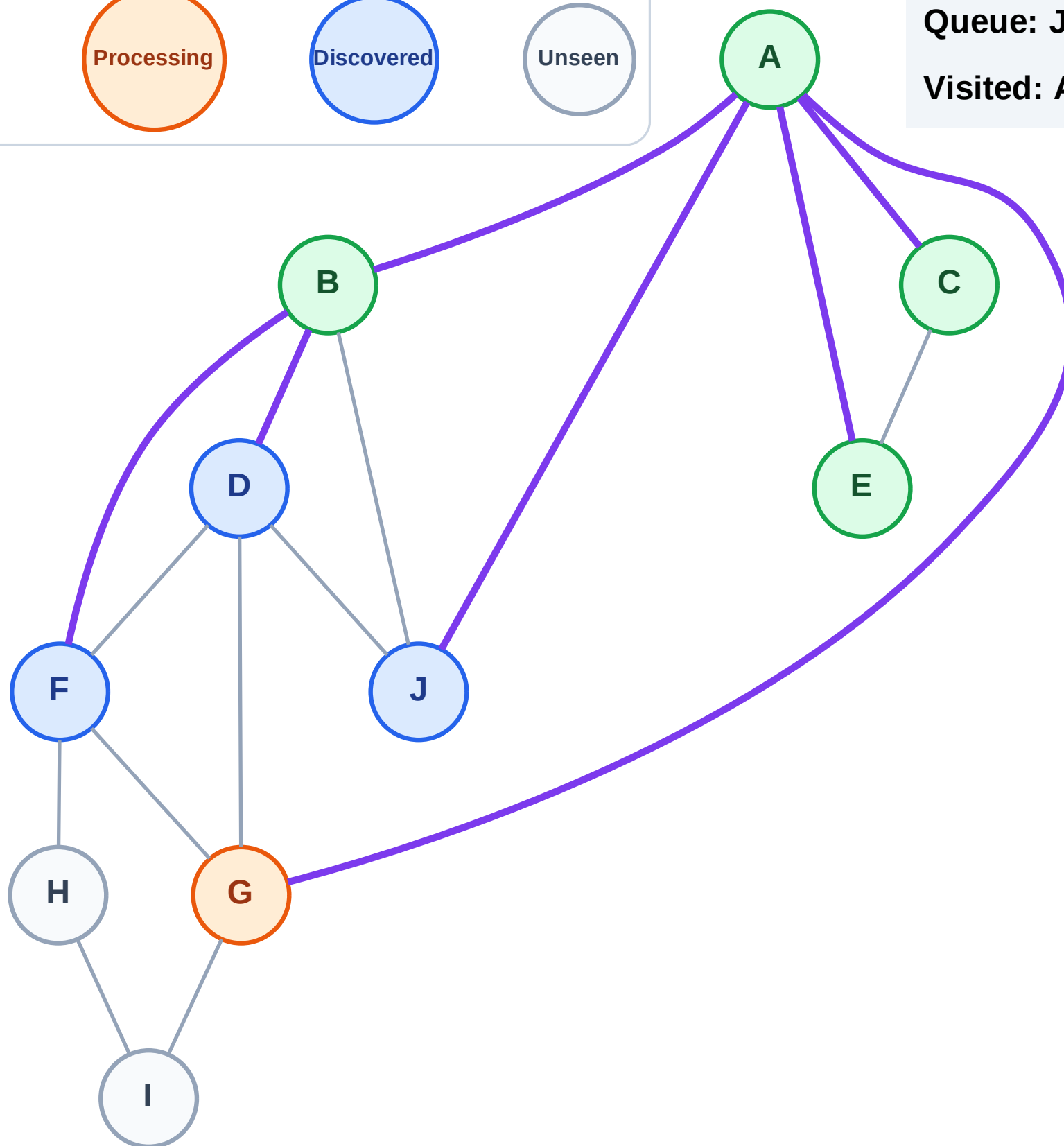
Step 10: Process node G

Legend



Queue: J → D → F

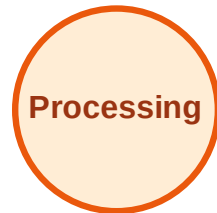
Visited: A, B, C, E



BFS Traversal

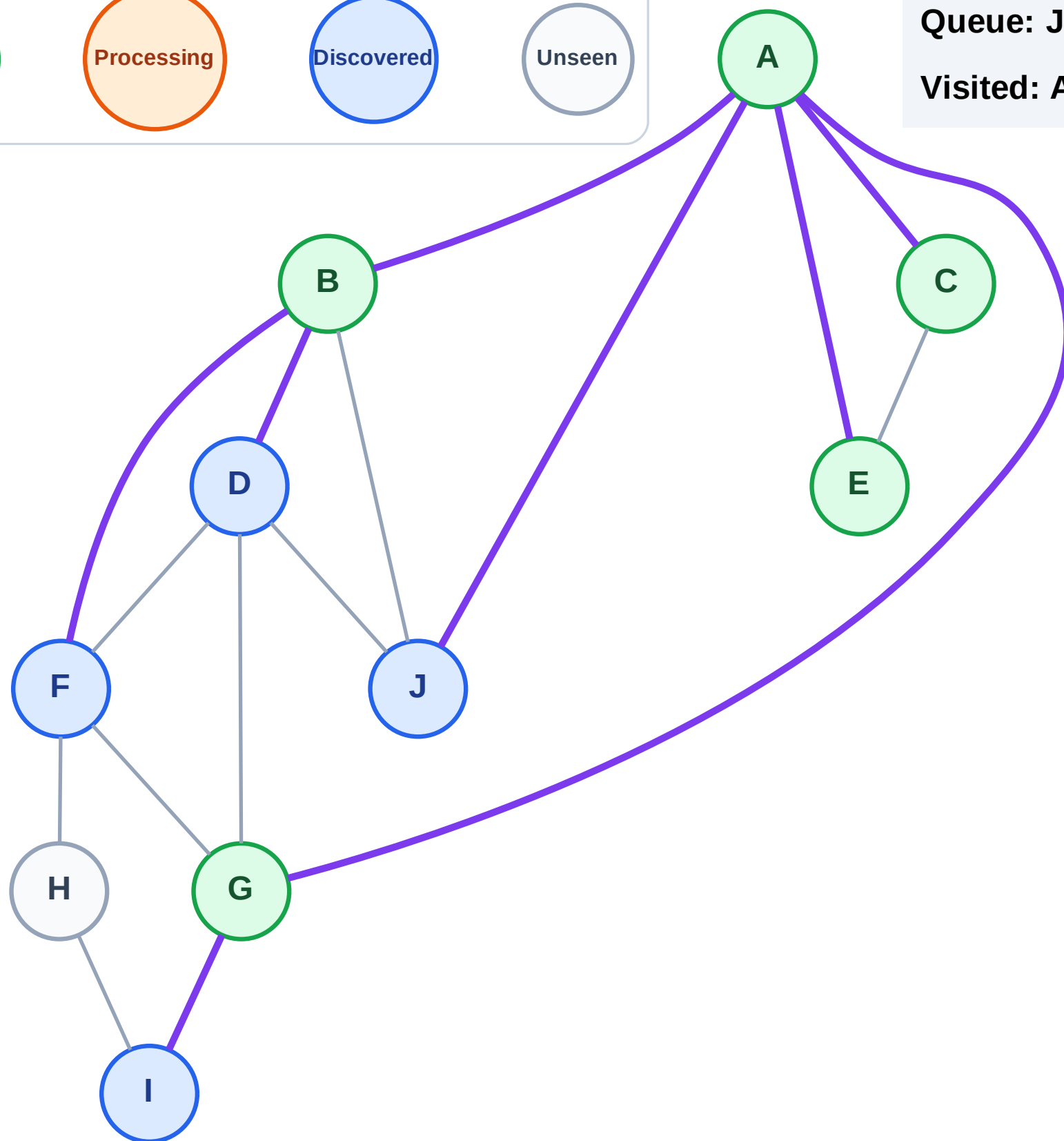
Step 11: Discover I from G

Legend



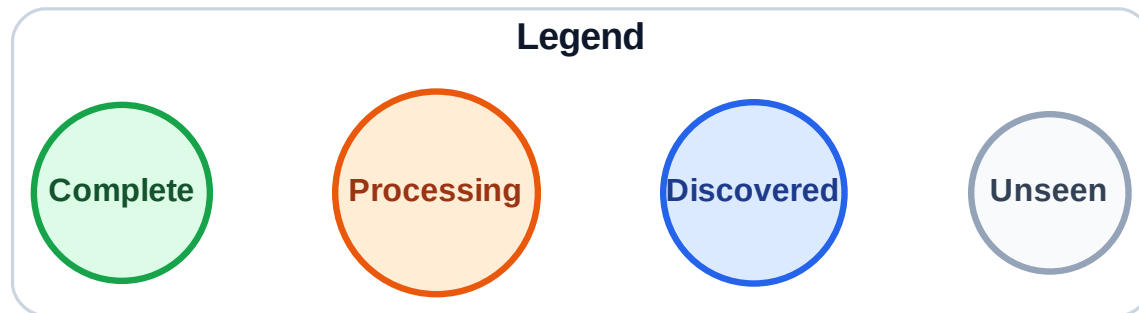
Queue: J → D → F → I

Visited: A, B, C, E, G

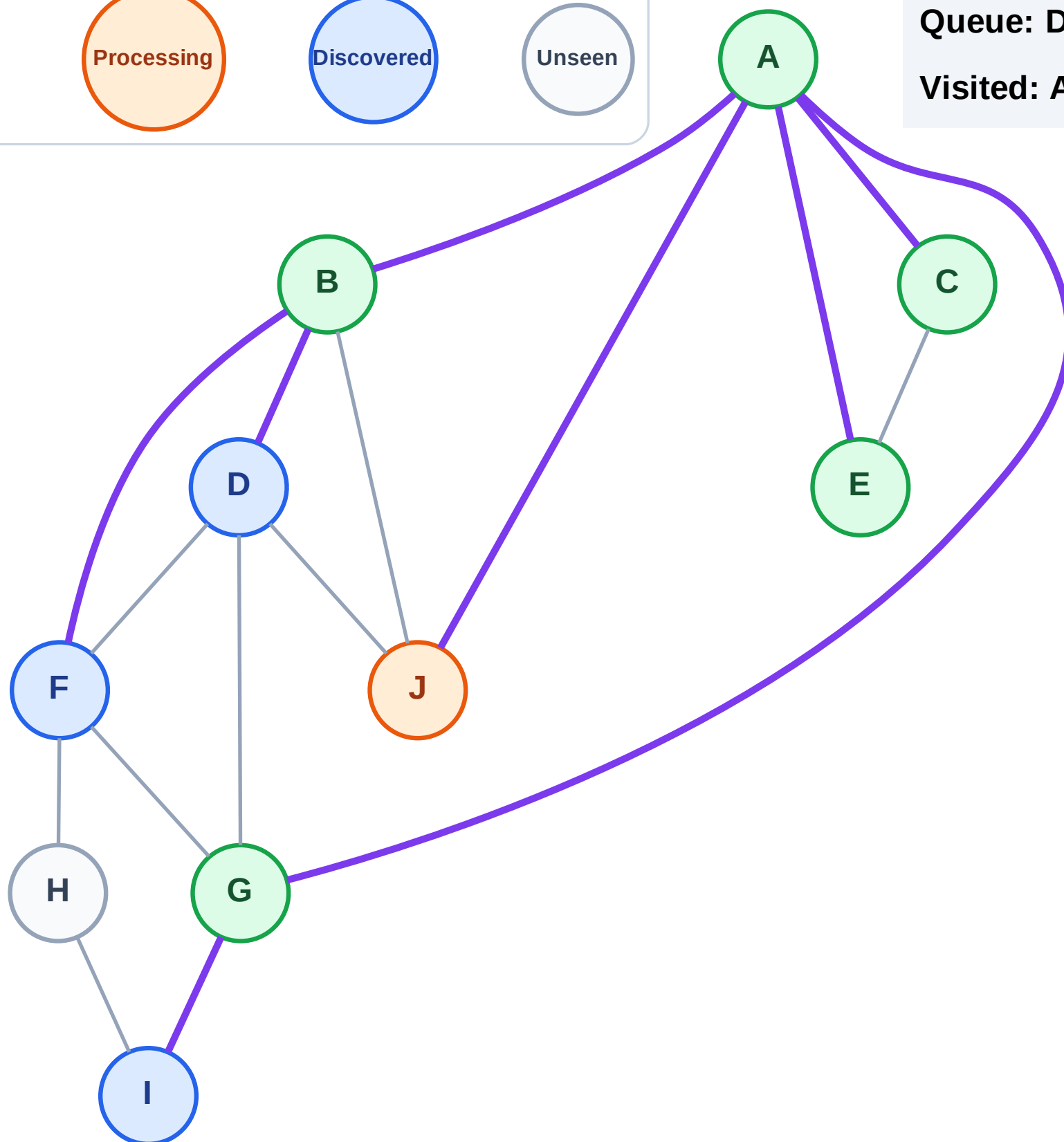


BFS Traversal

Step 12: Process node J



Queue: D → F → I
Visited: A, B, C, E, G



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

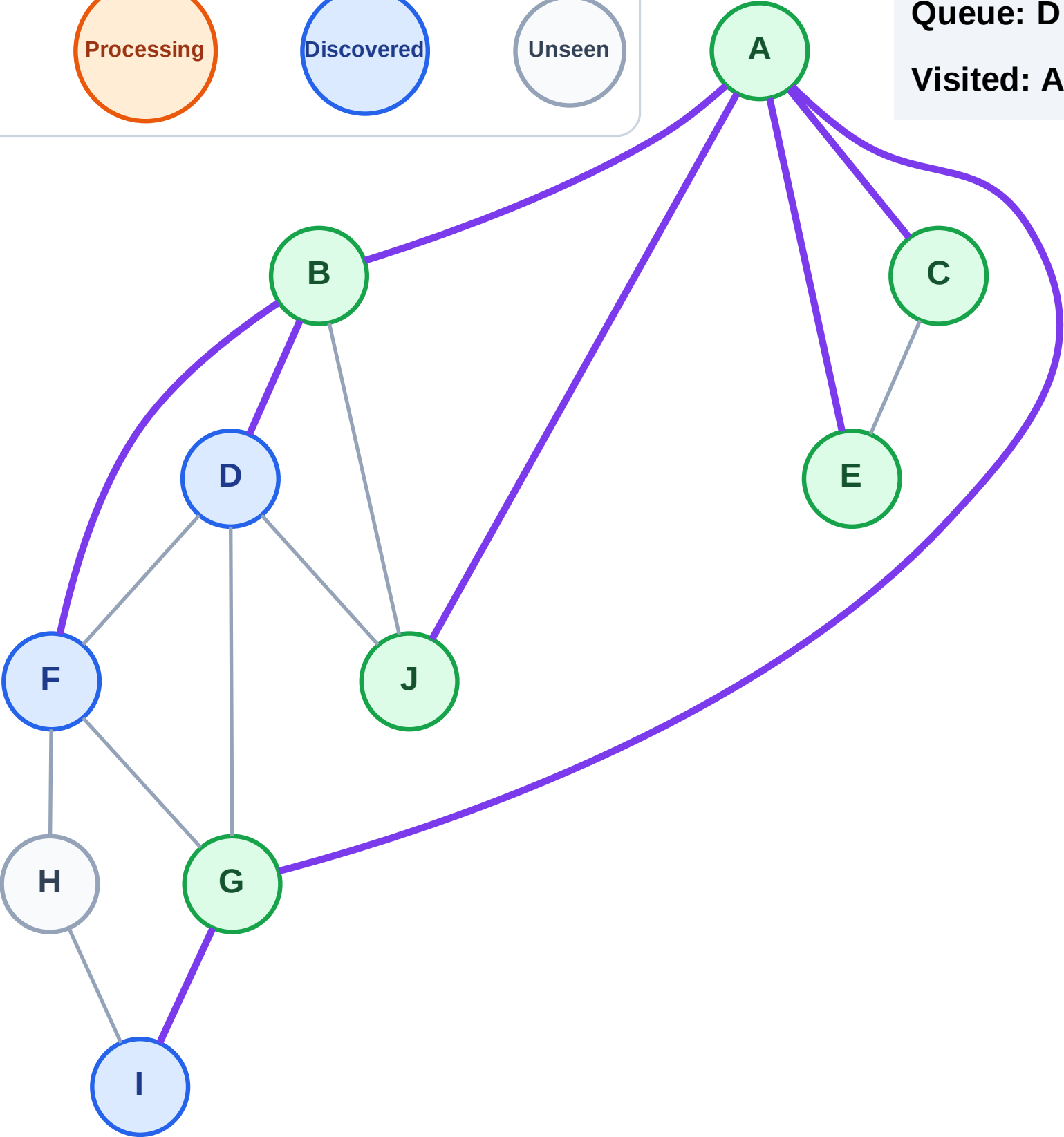
Processing

Discovered

Unseen

Queue: D → F → I

Visited: A, B, C, E, G, J



BFS Traversal

Step 14: Process node D

Legend

Complete

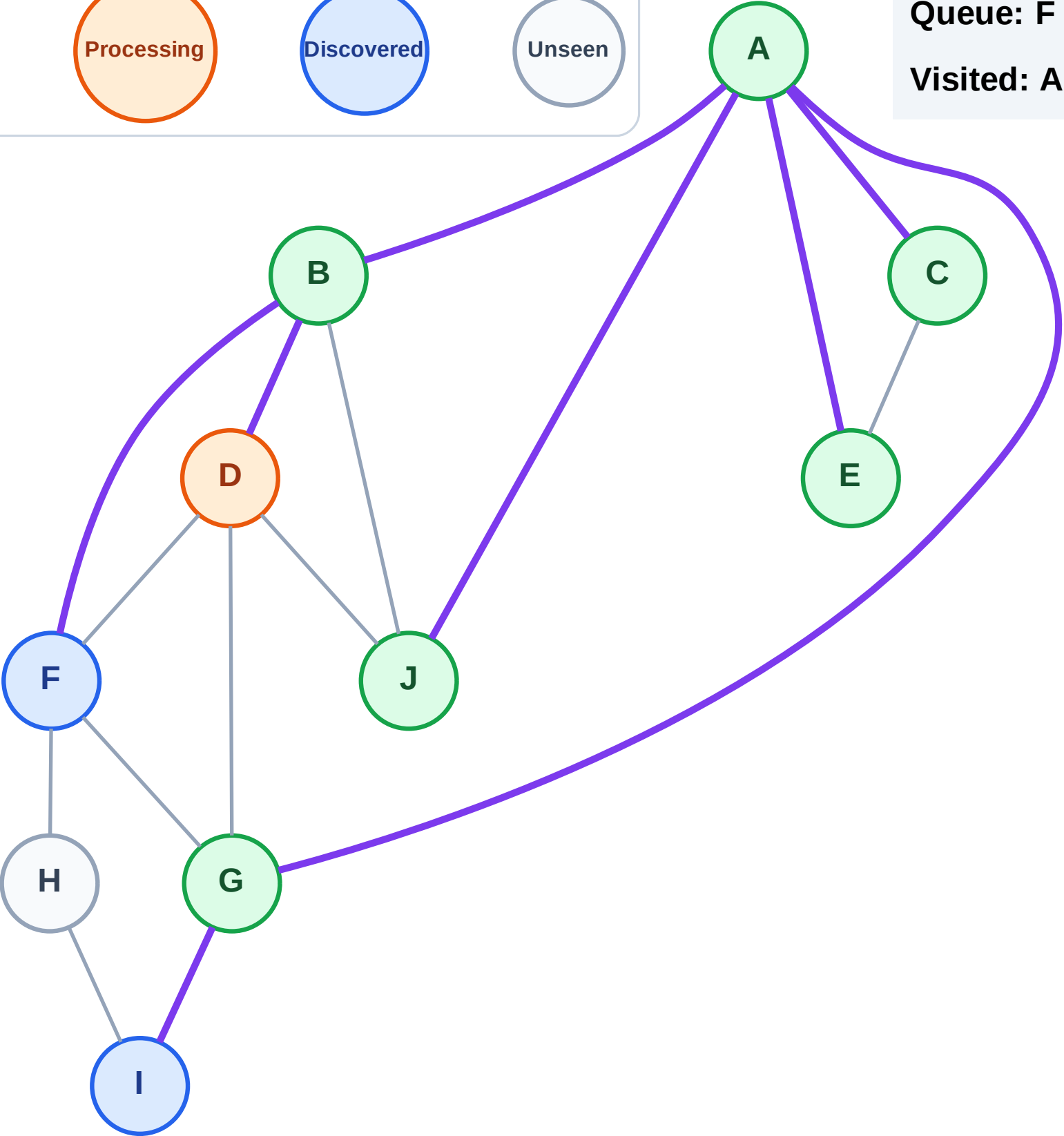
Processing

Discovered

Unseen

Queue: F → I

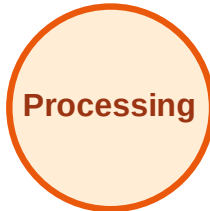
Visited: A, B, C, E, G, J



BFS Traversal

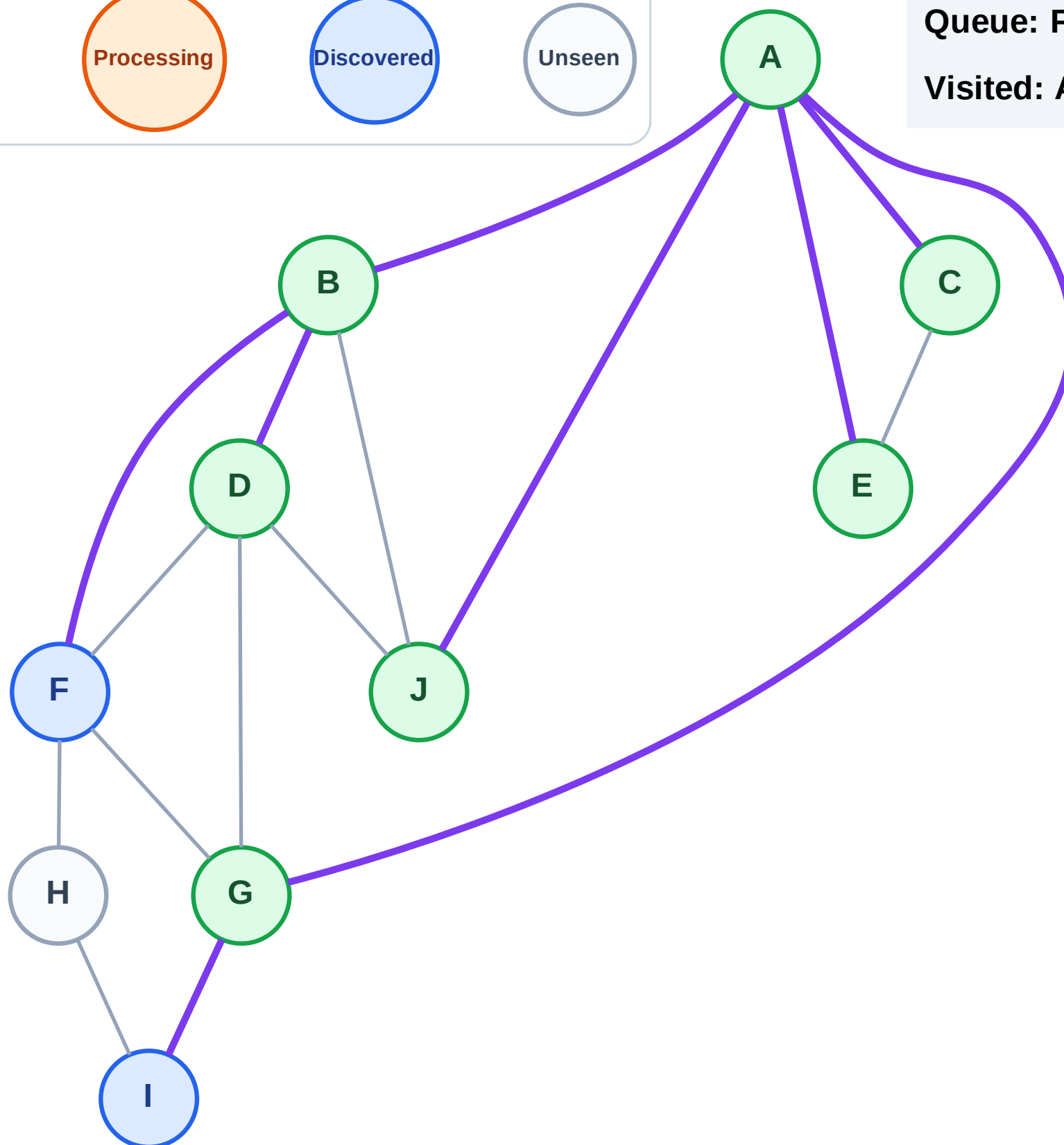
Step 15: No new neighbors from D

Legend



Queue: F → I

Visited: A, B, C, D, E, G, J



BFS Traversal

Step 16: Process node F

Legend

Complete

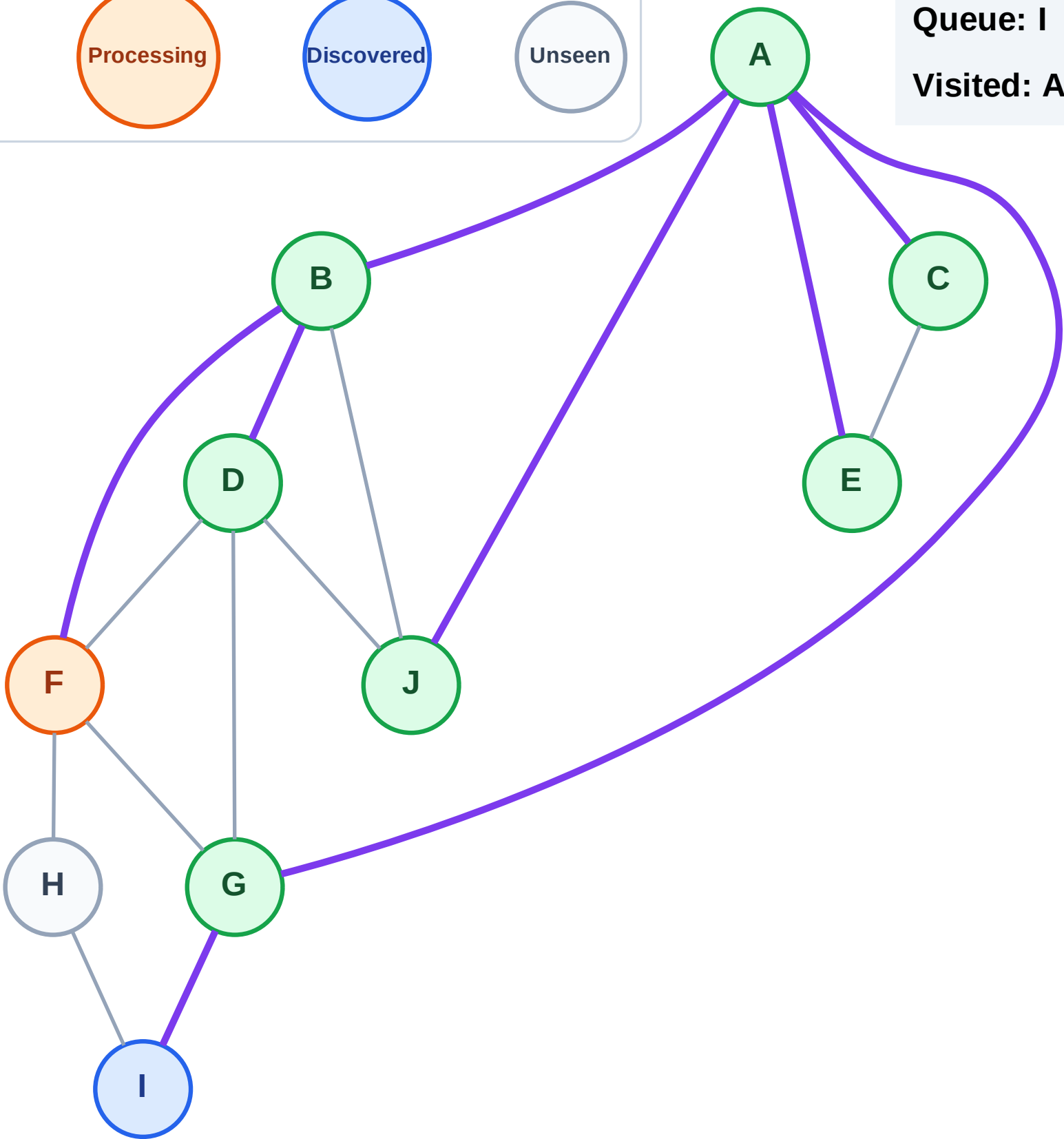
Processing

Discovered

Unseen

Queue: I

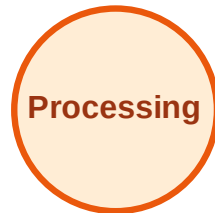
Visited: A, B, C, D, E, G, J



BFS Traversal

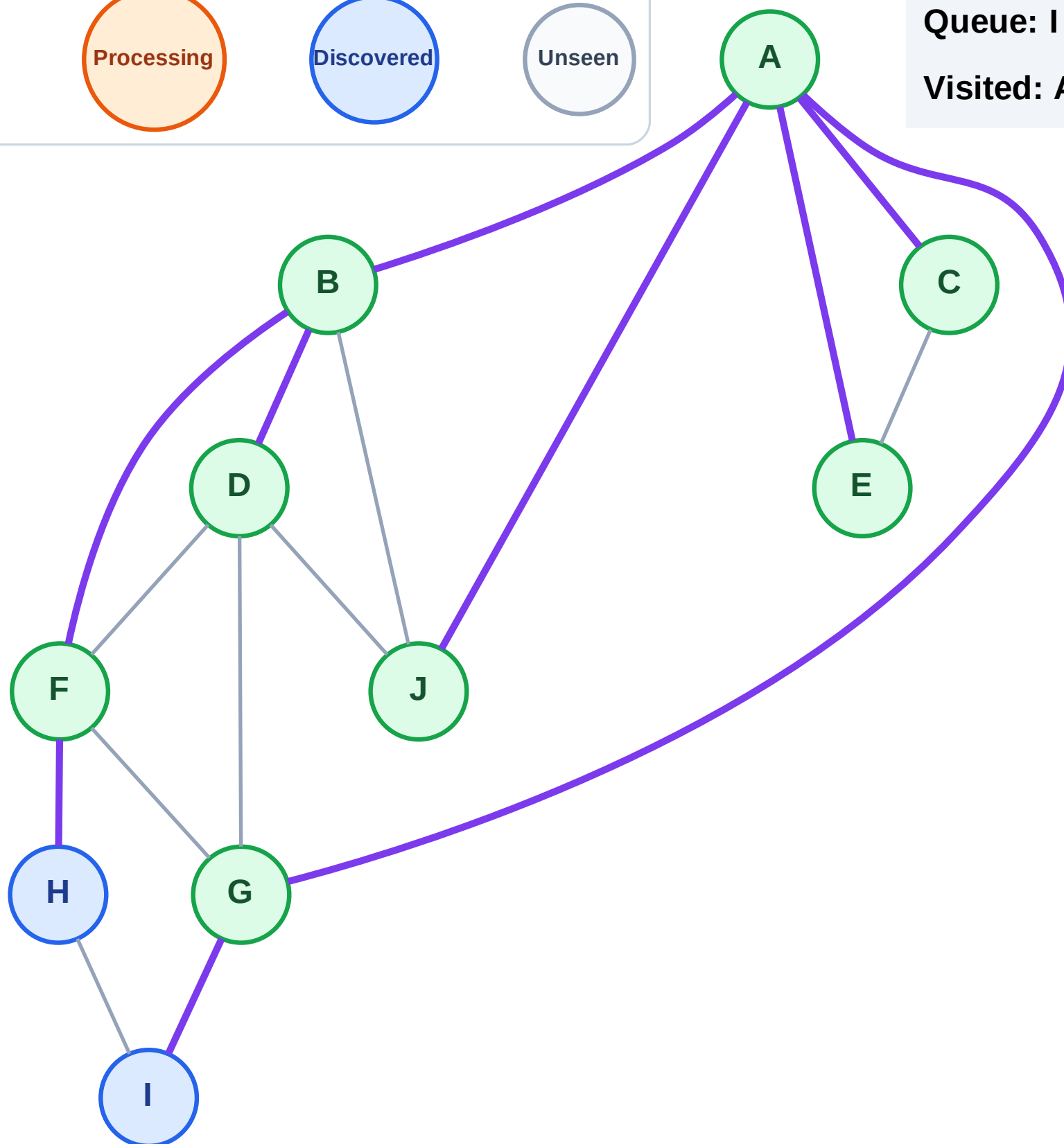
Step 17: Discover H from F

Legend



Queue: I → H

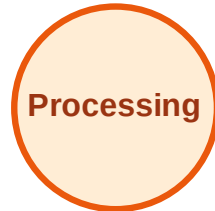
Visited: A, B, C, D, E, F, G, J



BFS Traversal

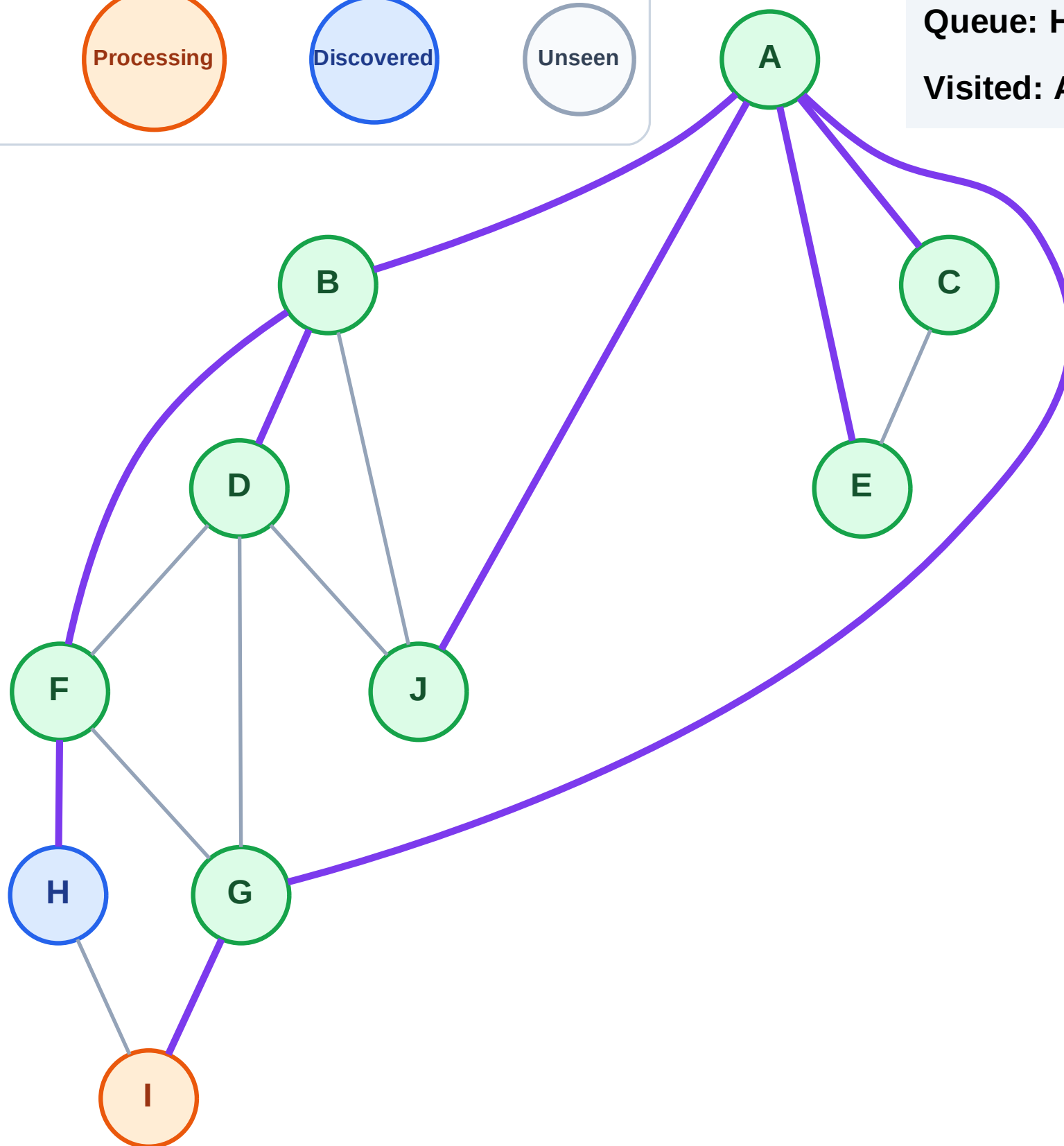
Step 18: Process node I

Legend



Queue: H

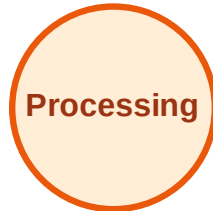
Visited: A, B, C, D, E, F, G, J



BFS Traversal

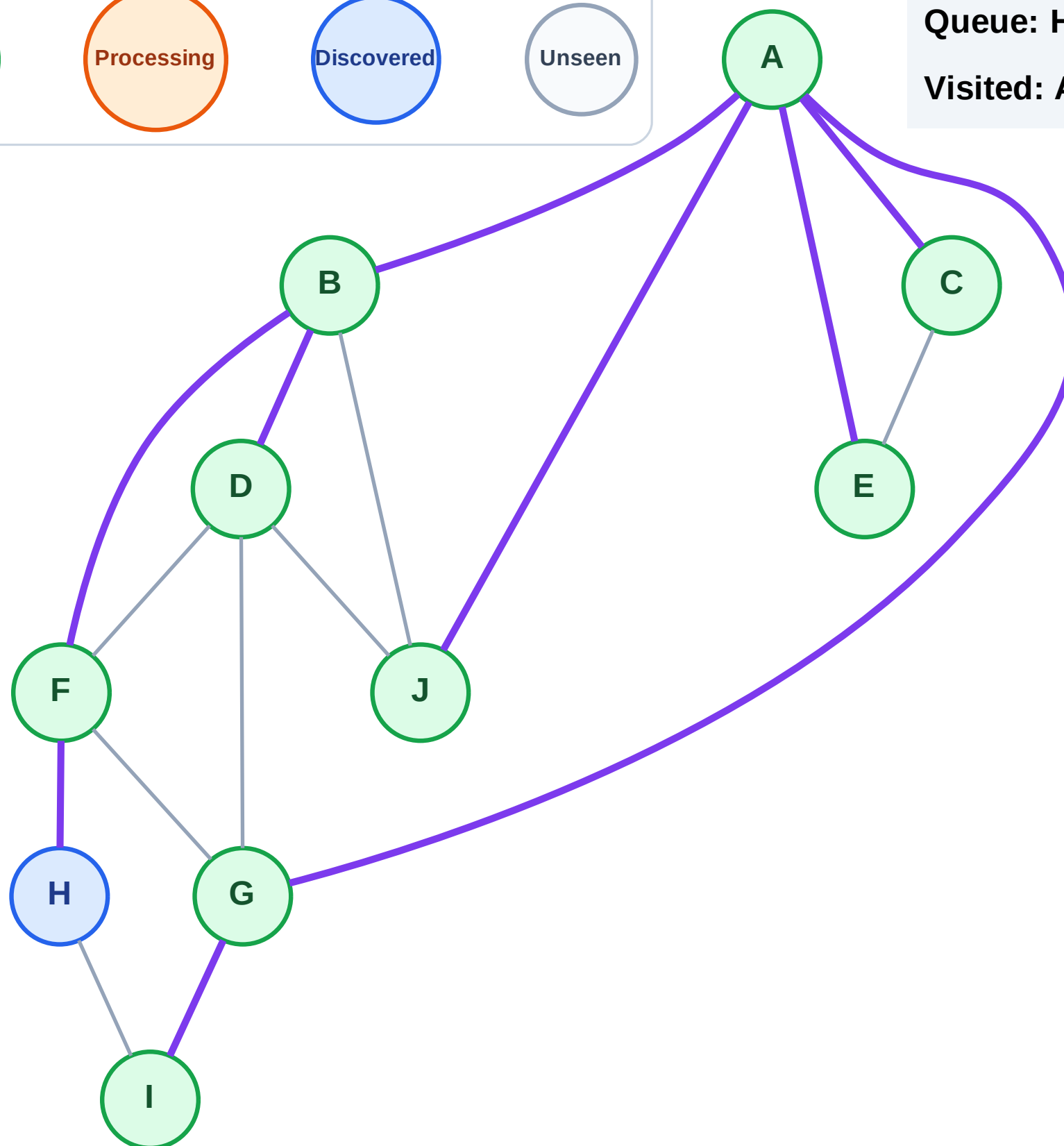
Step 19: No new neighbors from I

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

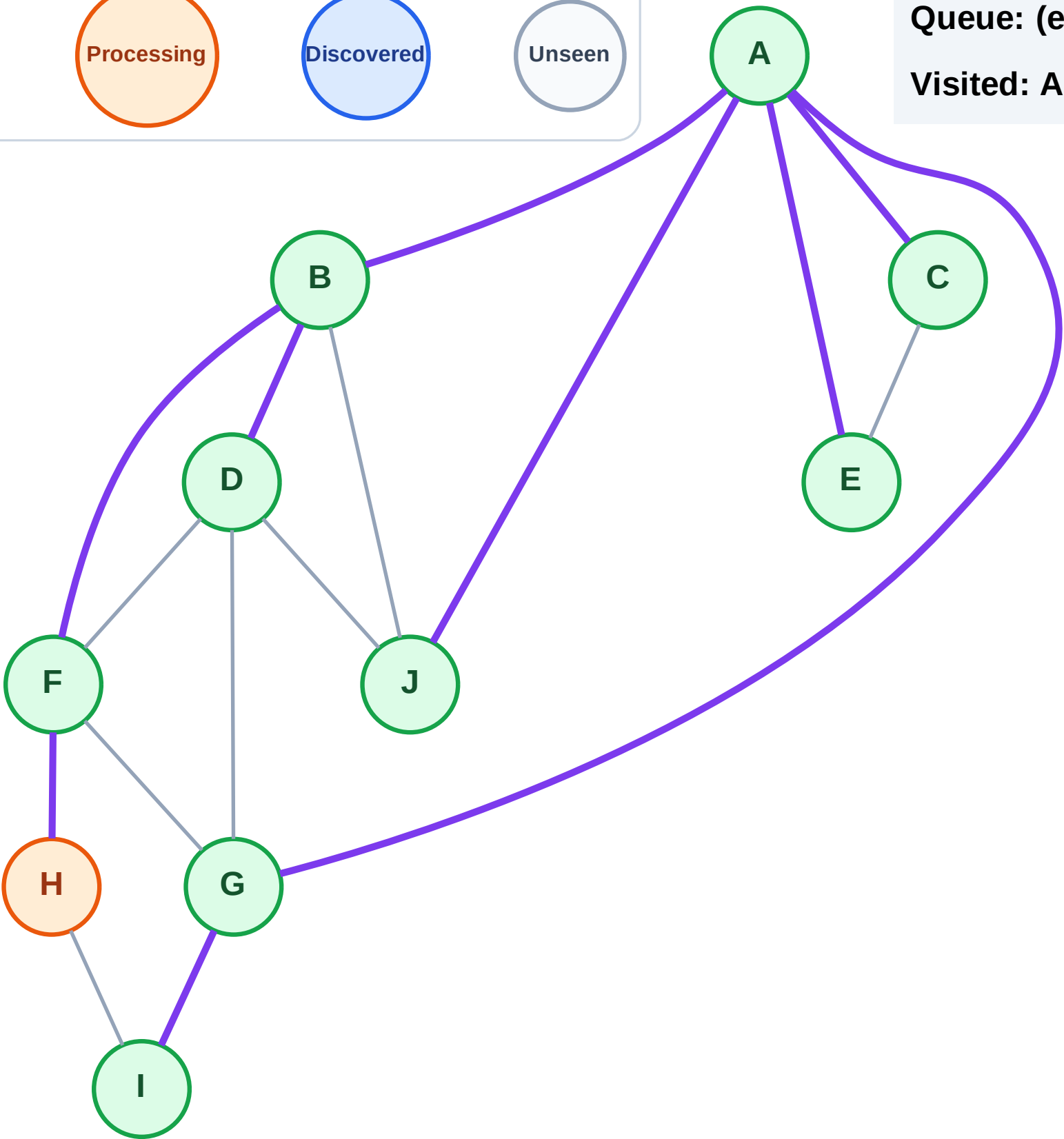
Complete

Processing

Discovered

Unseen

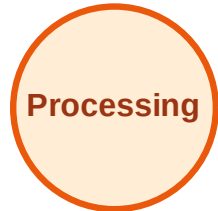
Queue: (empty)
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

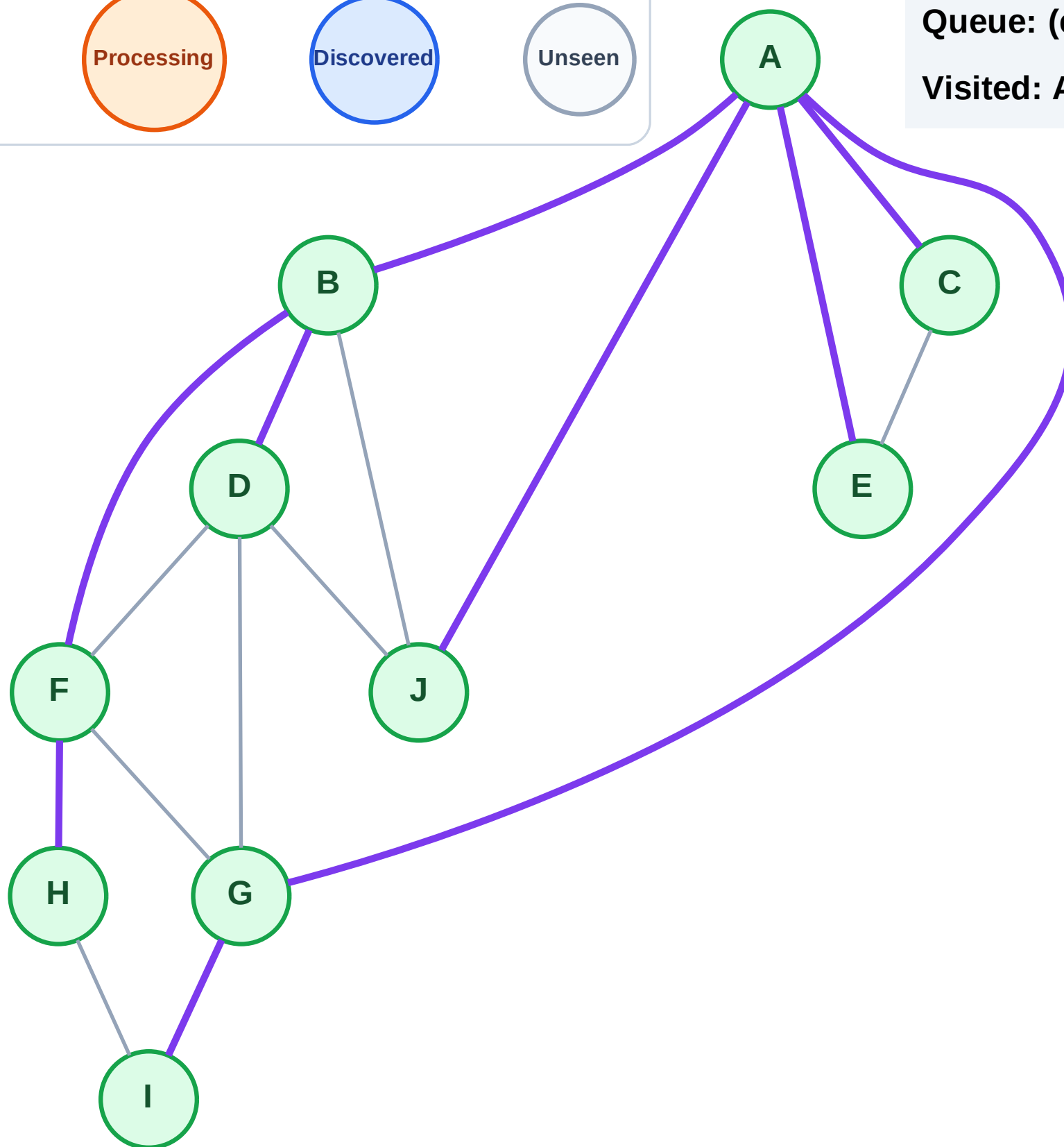
Step 21: No new neighbors from H

Legend



Queue: (empty)

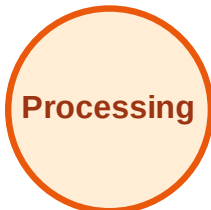
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

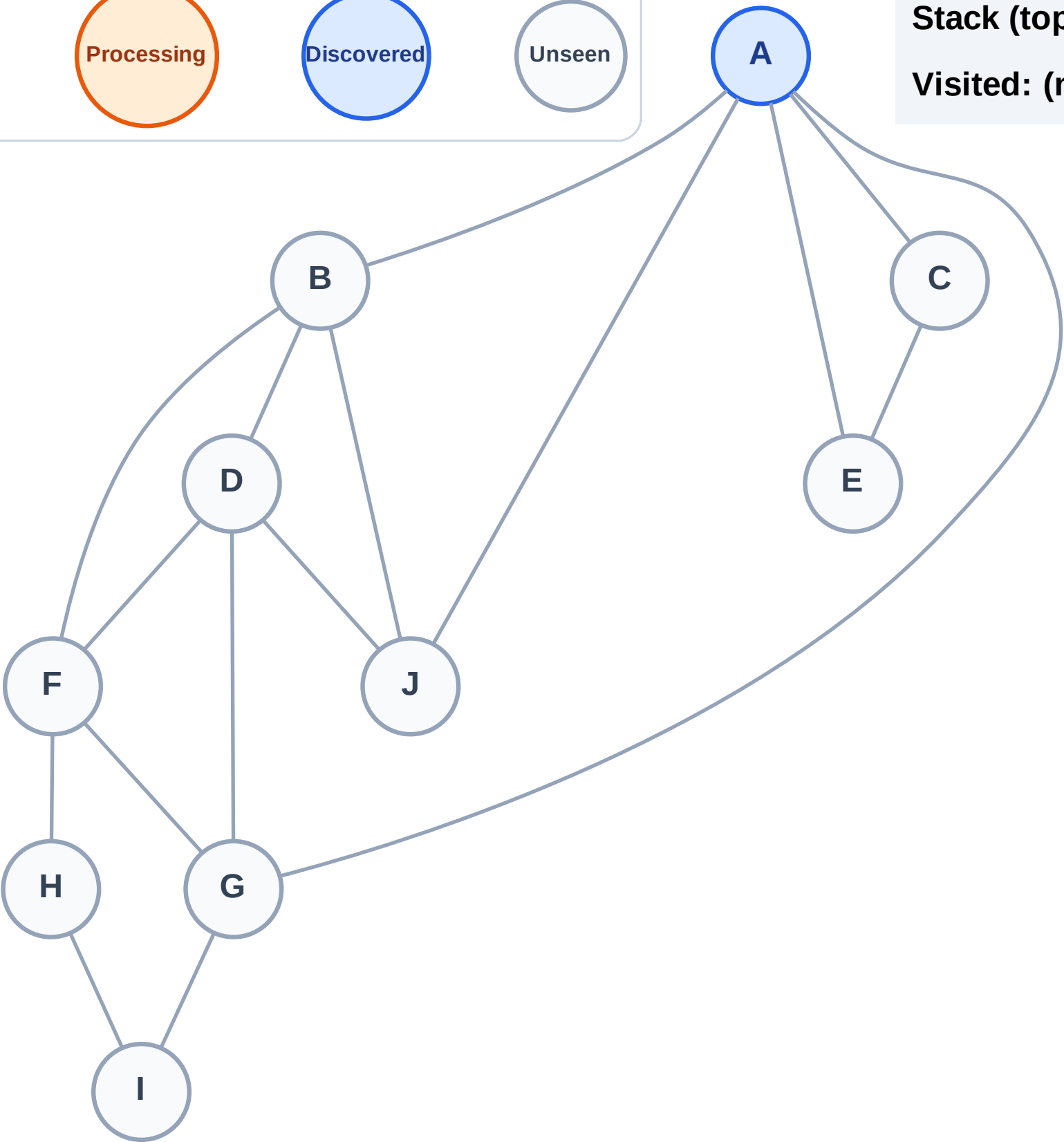
Step 1: Start at A

Legend



Stack (top at right): A

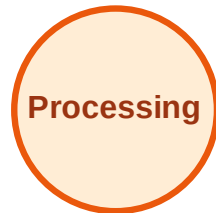
Visited: (none)



DFS Traversal

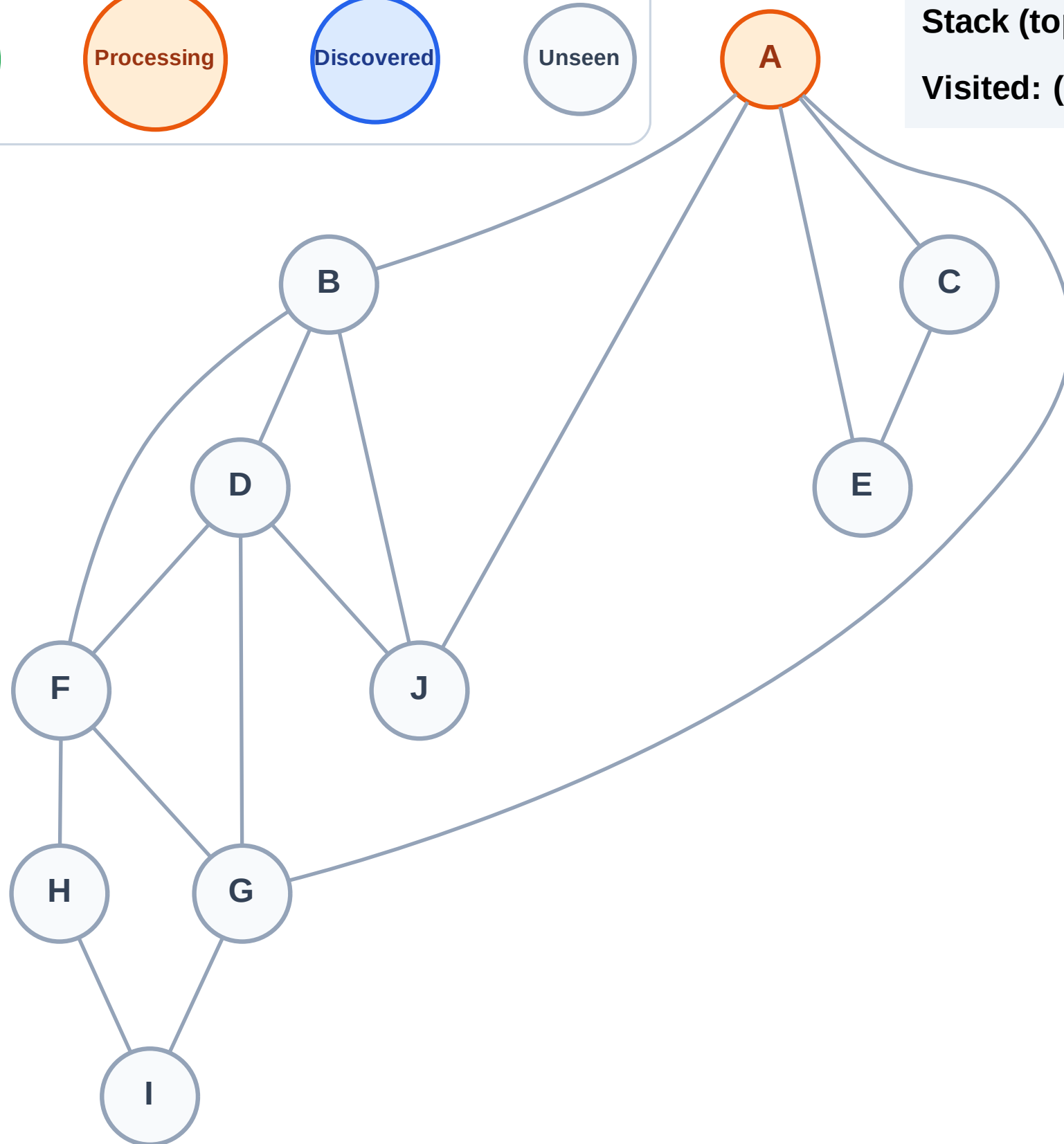
Step 2: Process node A

Legend



Stack (top at right): (empty)

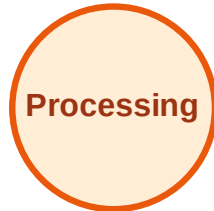
Visited: (none)



DFS Traversal

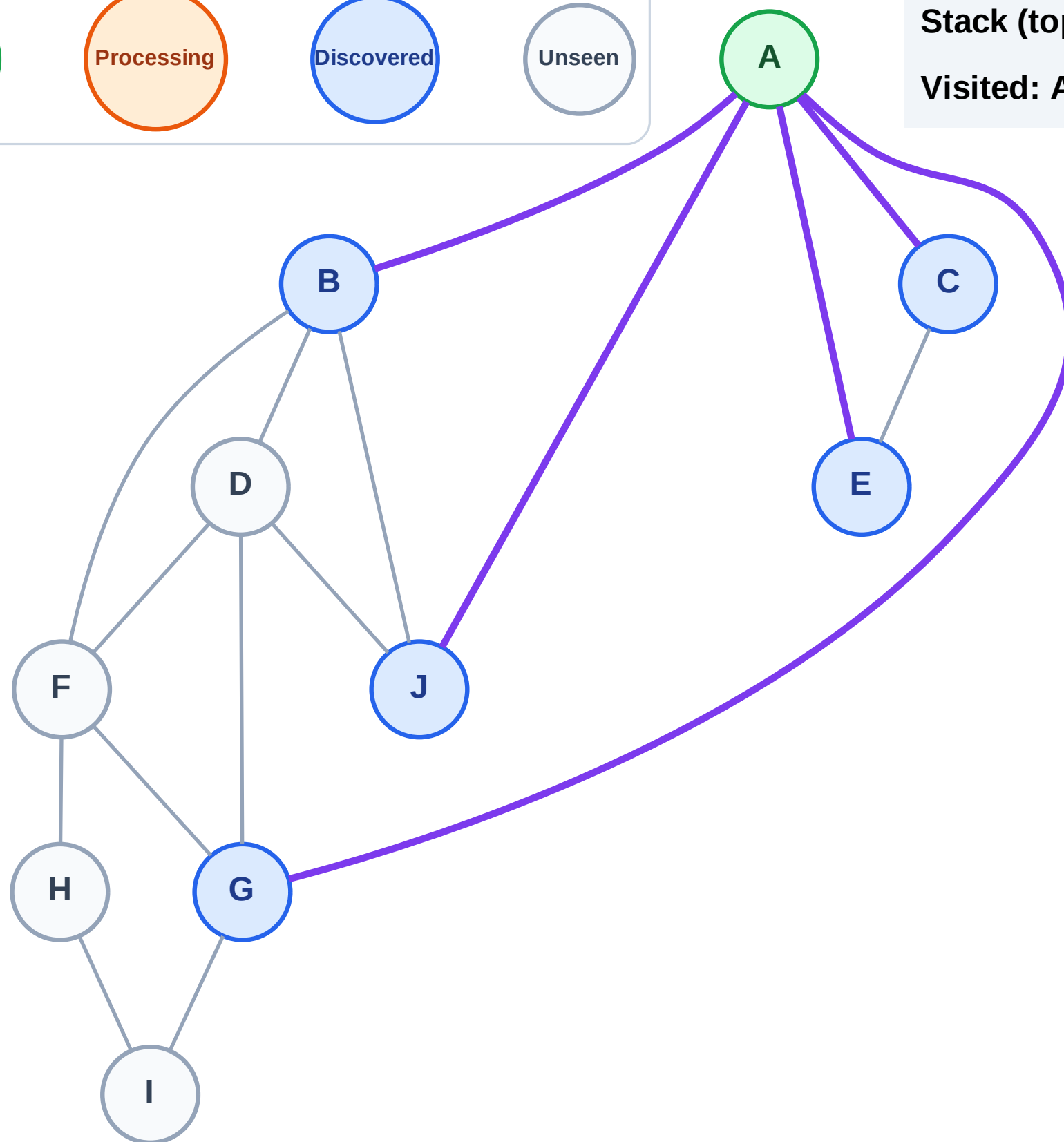
Step 3: Discover J, G, E, C, B from A

Legend



Stack (top at right): J | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

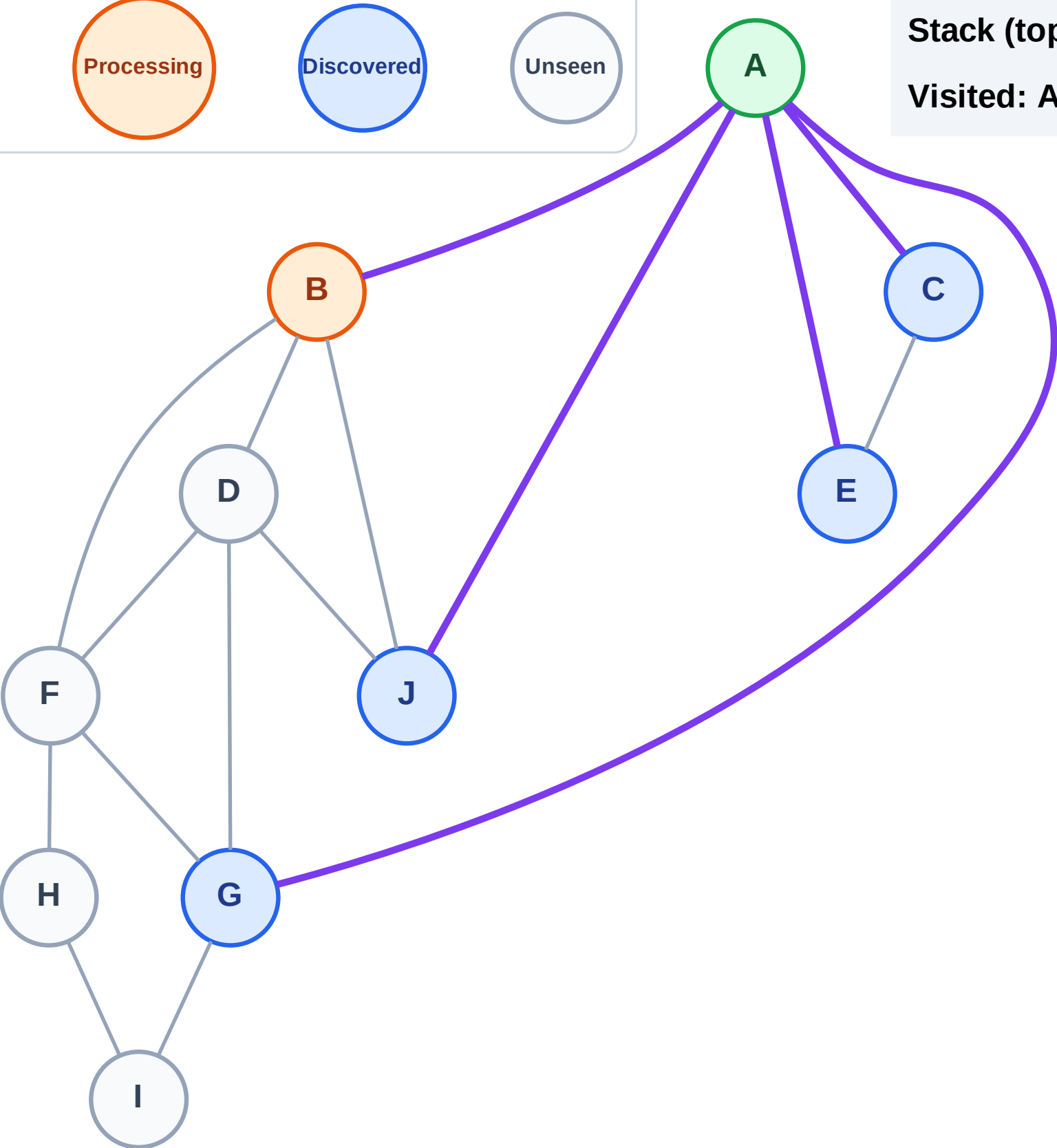
Complete

Processing

Discovered

Unseen

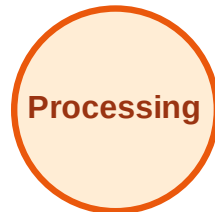
Stack (top at right): J | G | E | C
Visited: A



DFS Traversal

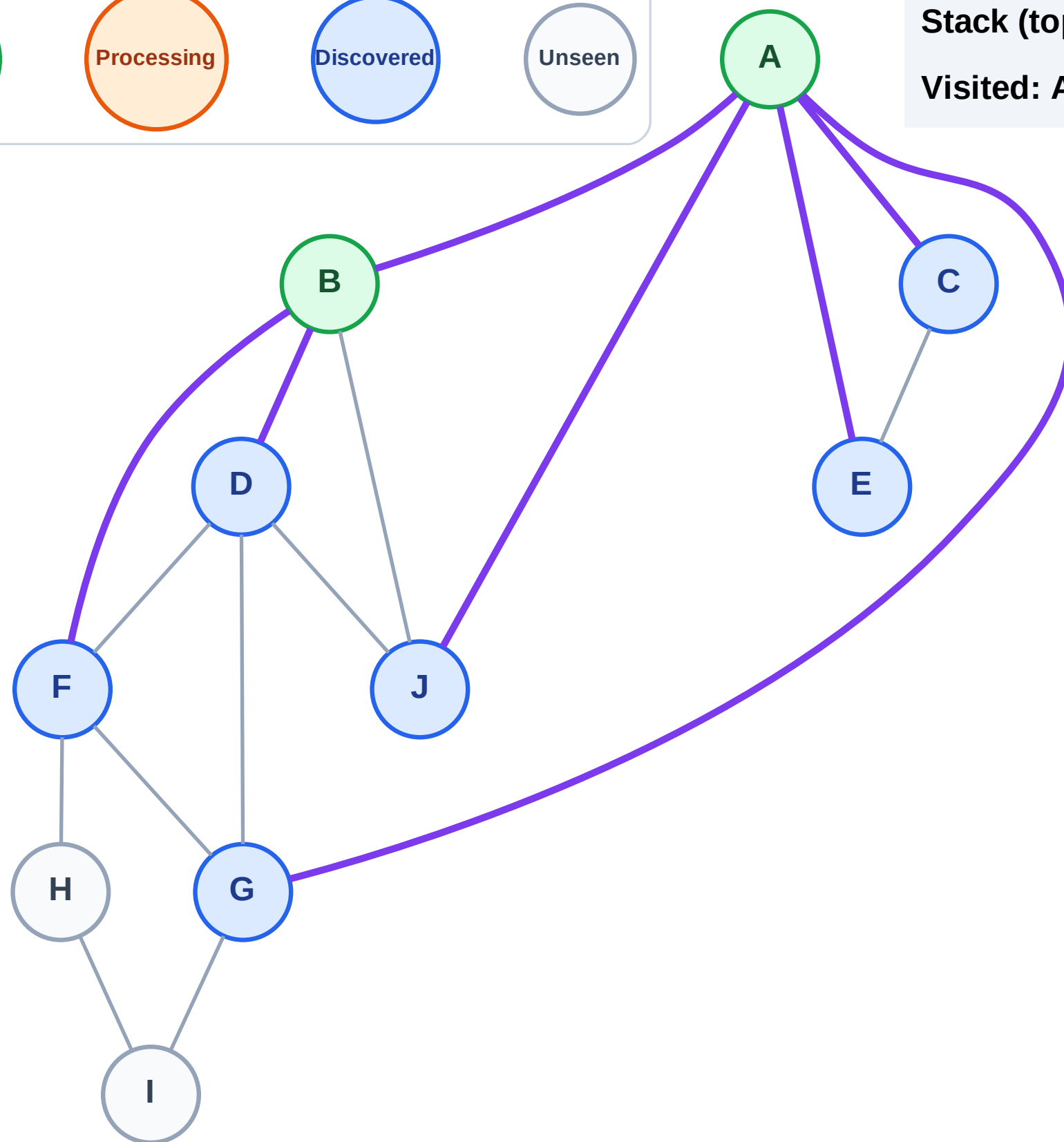
Step 5: Discover F, D from B

Legend



Stack (top at right): J | G | E | C | F | D

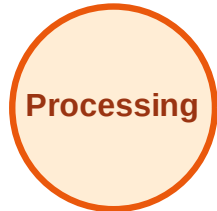
Visited: A, B



DFS Traversal

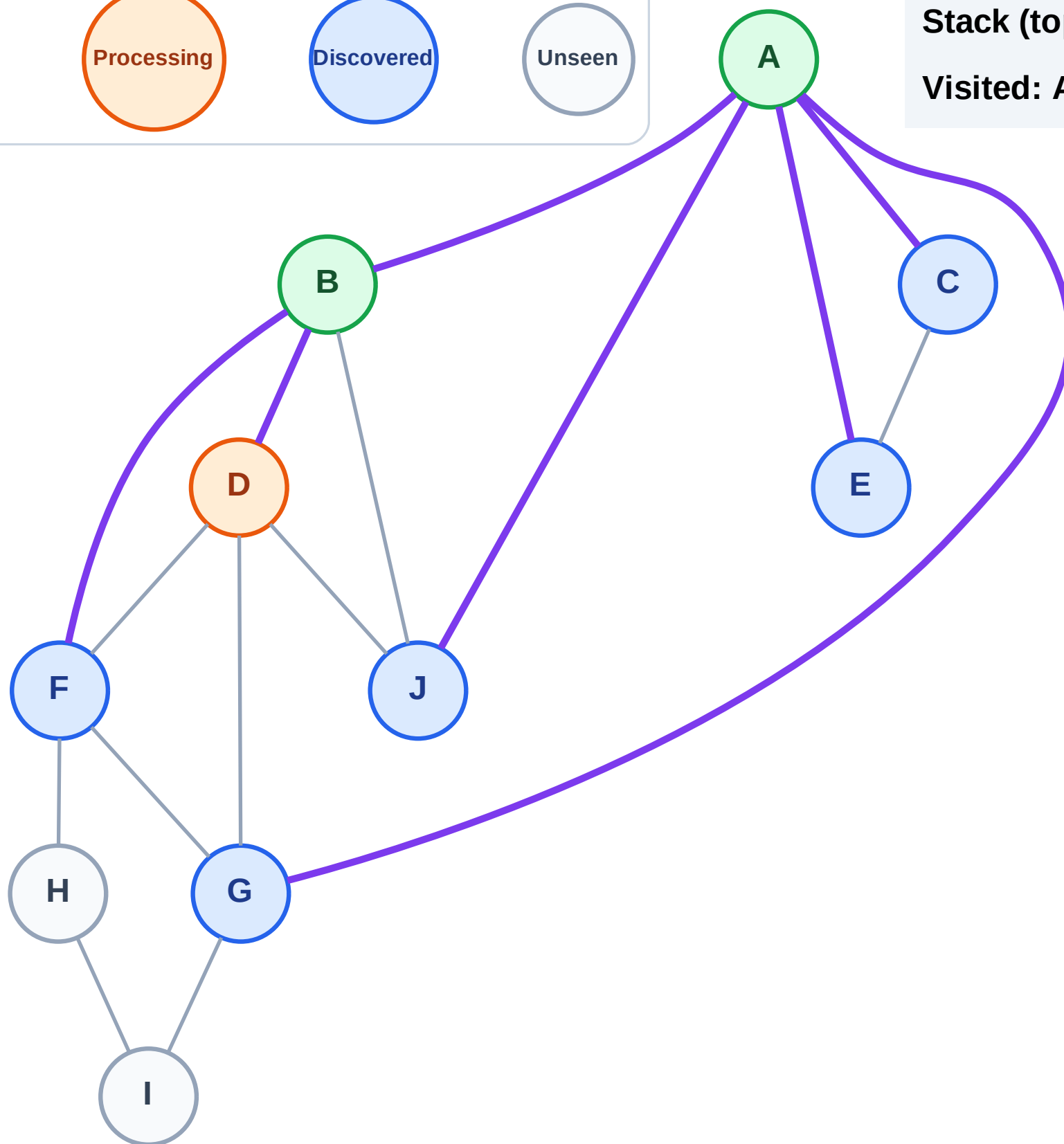
Step 6: Process node D

Legend



Stack (top at right): J | G | E | C | F

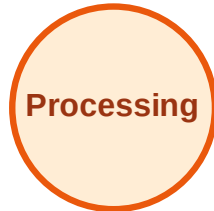
Visited: A, B



DFS Traversal

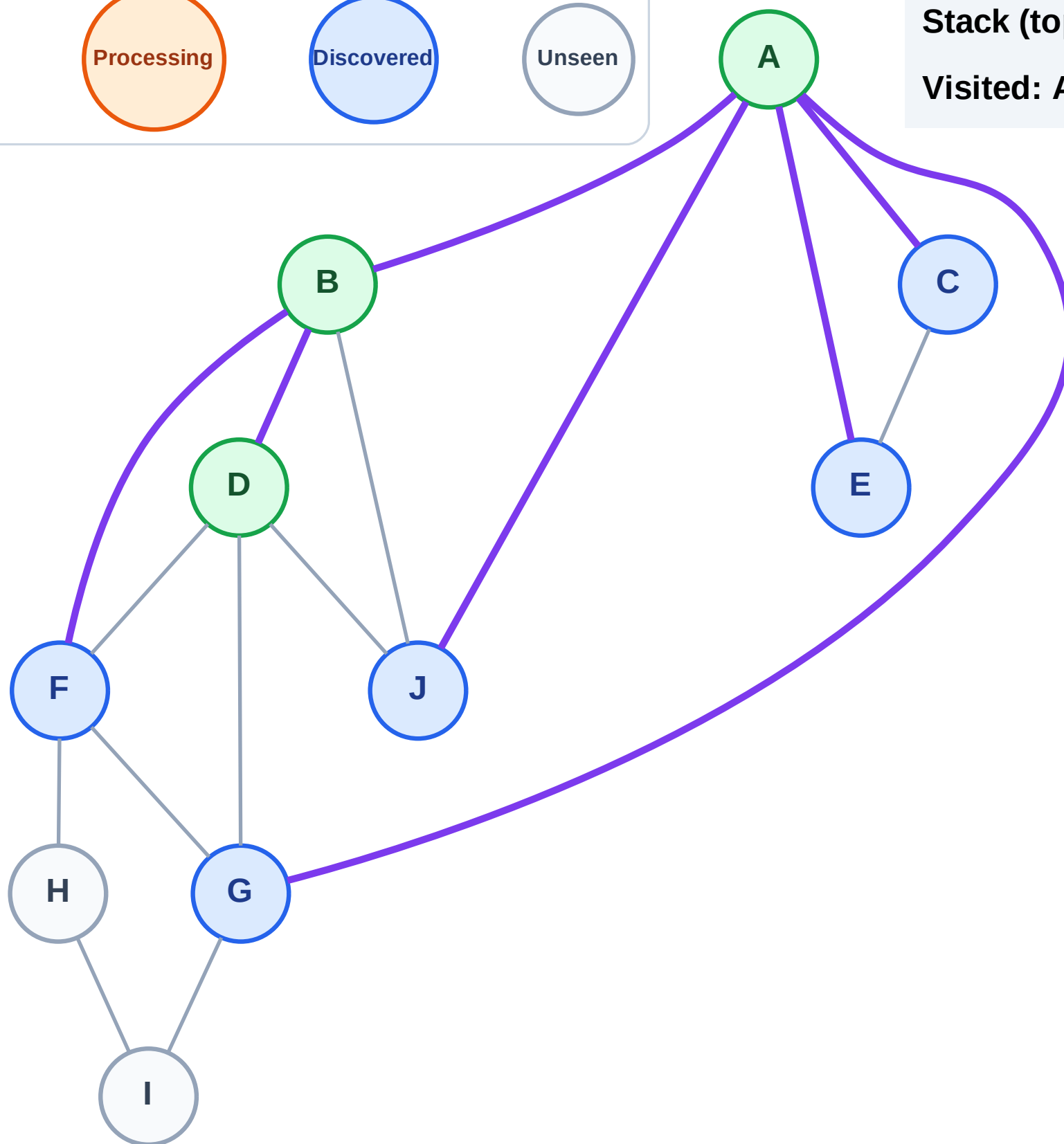
Step 7: No new neighbors from D

Legend



Stack (top at right): J | G | E | C | F

Visited: A, B, D



DFS Traversal

Step 8: Process node F

Legend

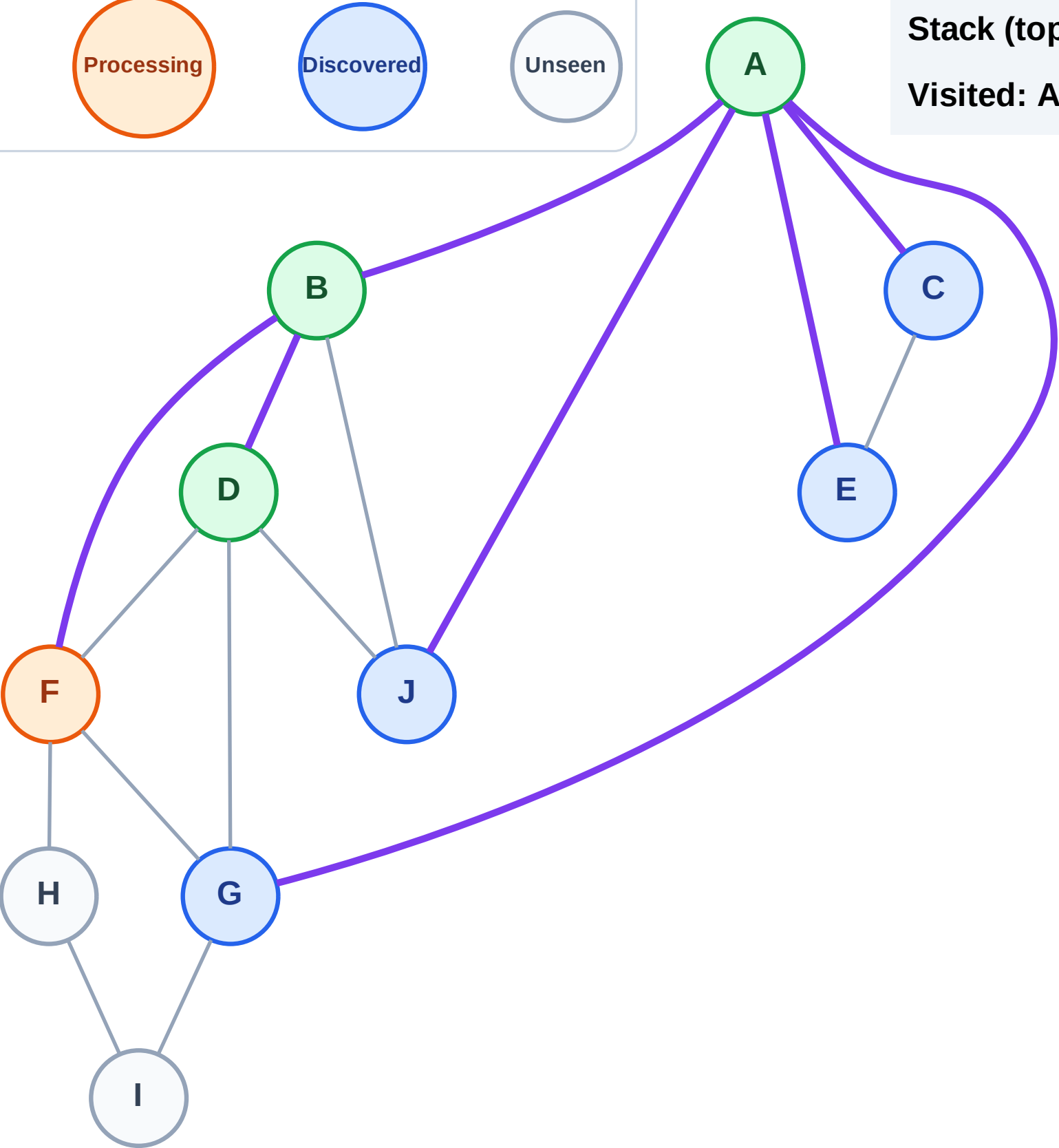
Complete

Processing

Discovered

Unseen

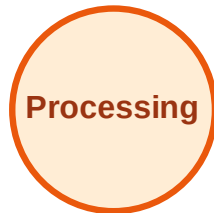
Stack (top at right): J | G | E | C
Visited: A, B, D



DFS Traversal

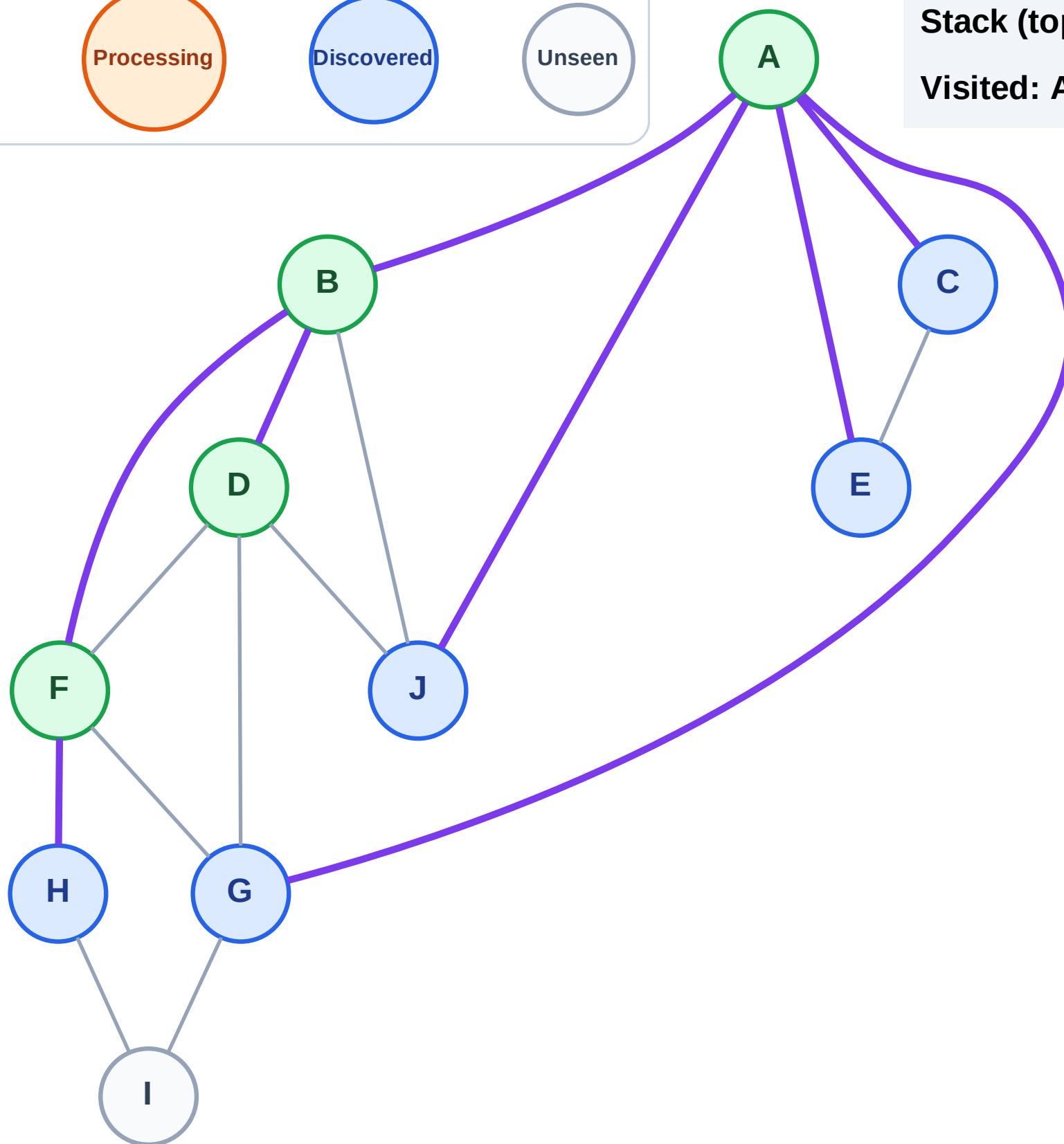
Step 9: Discover H from F

Legend



Stack (top at right): J | G | E | C | H

Visited: A, B, D, F



DFS Traversal

Step 10: Process node H

Legend

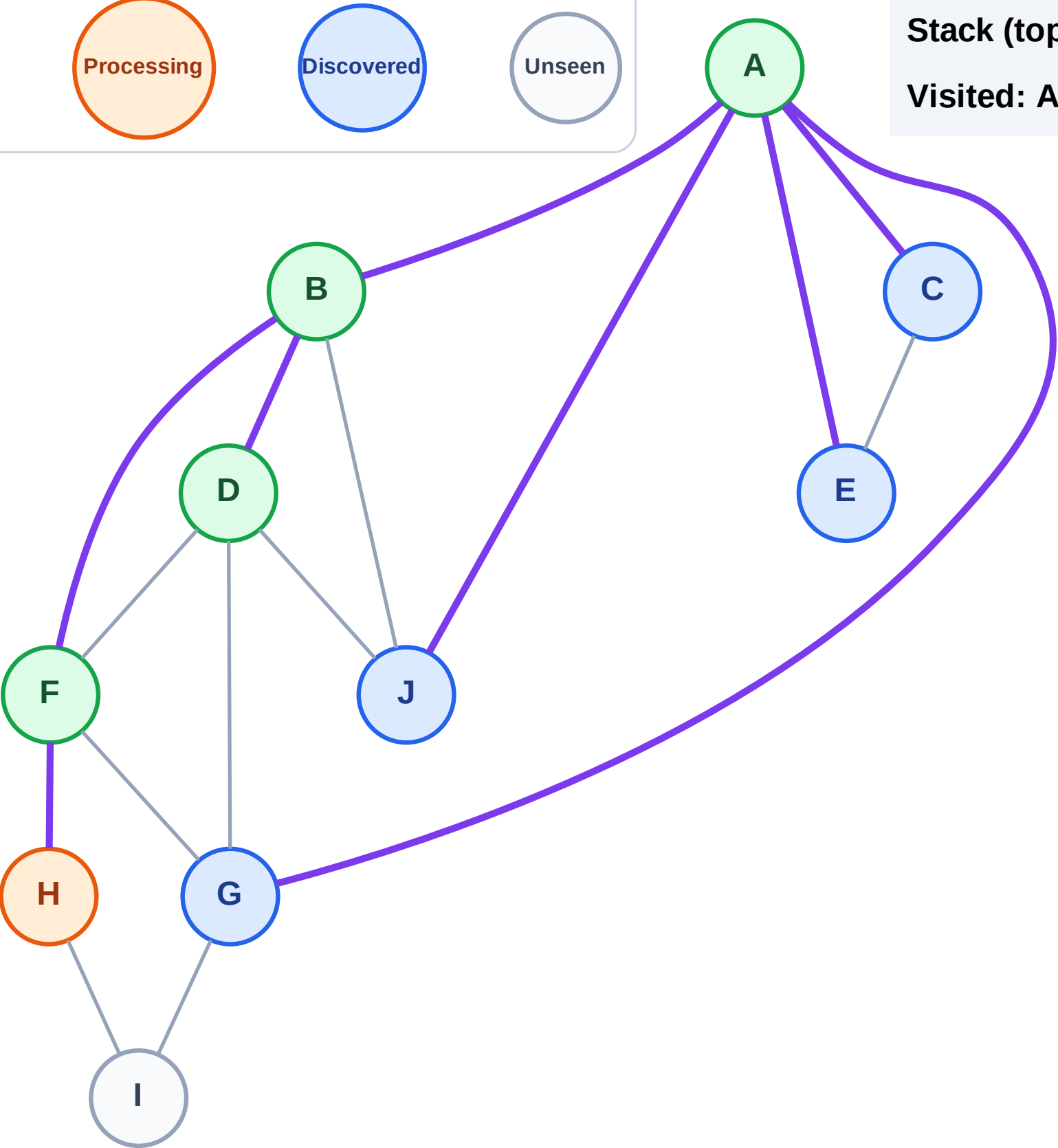
Complete

Processing

Discovered

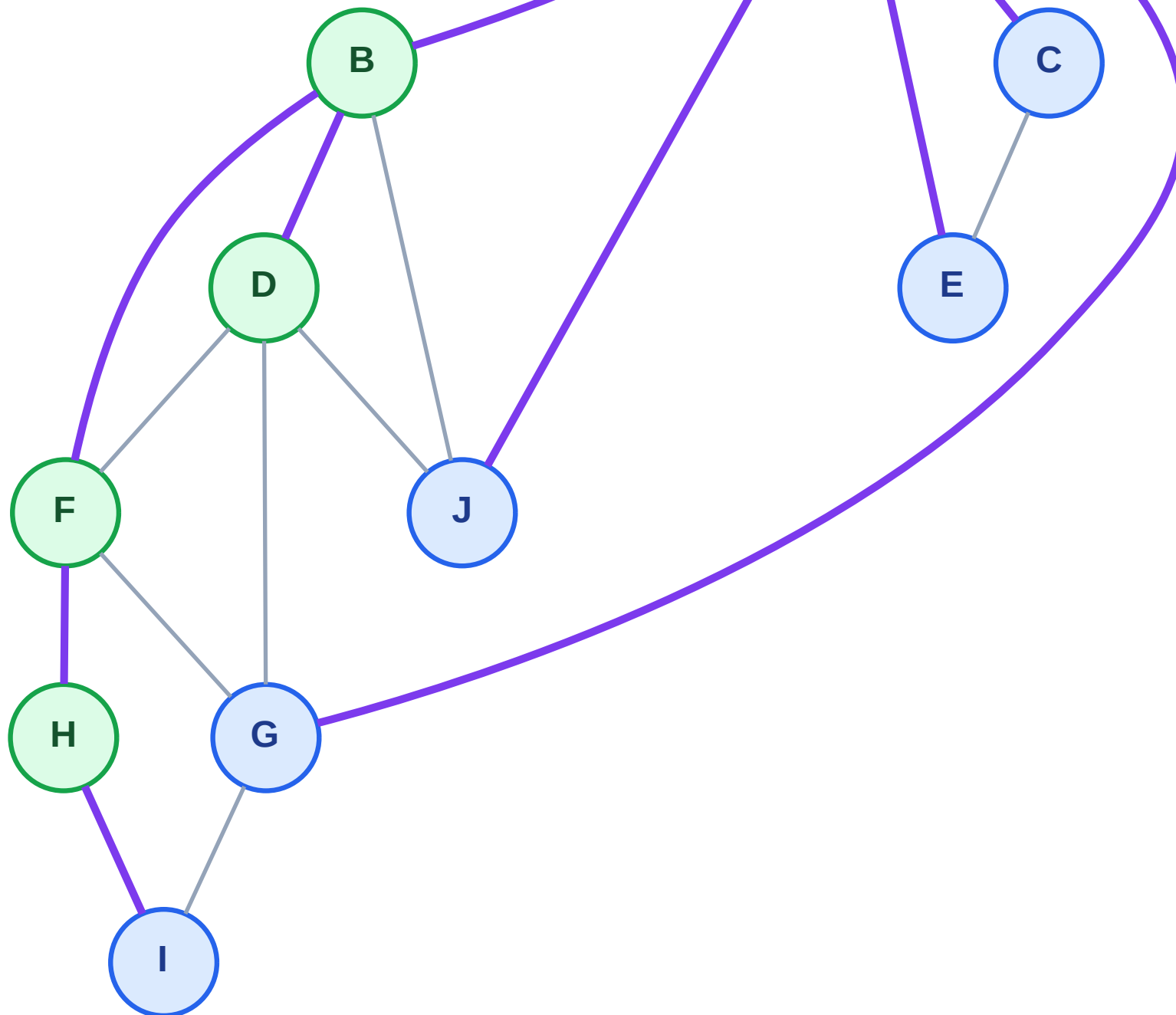
Unseen

Stack (top at right): J | G | E | C
Visited: A, B, D, F



Step 11: Discover I from H

Step 11: Discover I from H



DFS Traversal

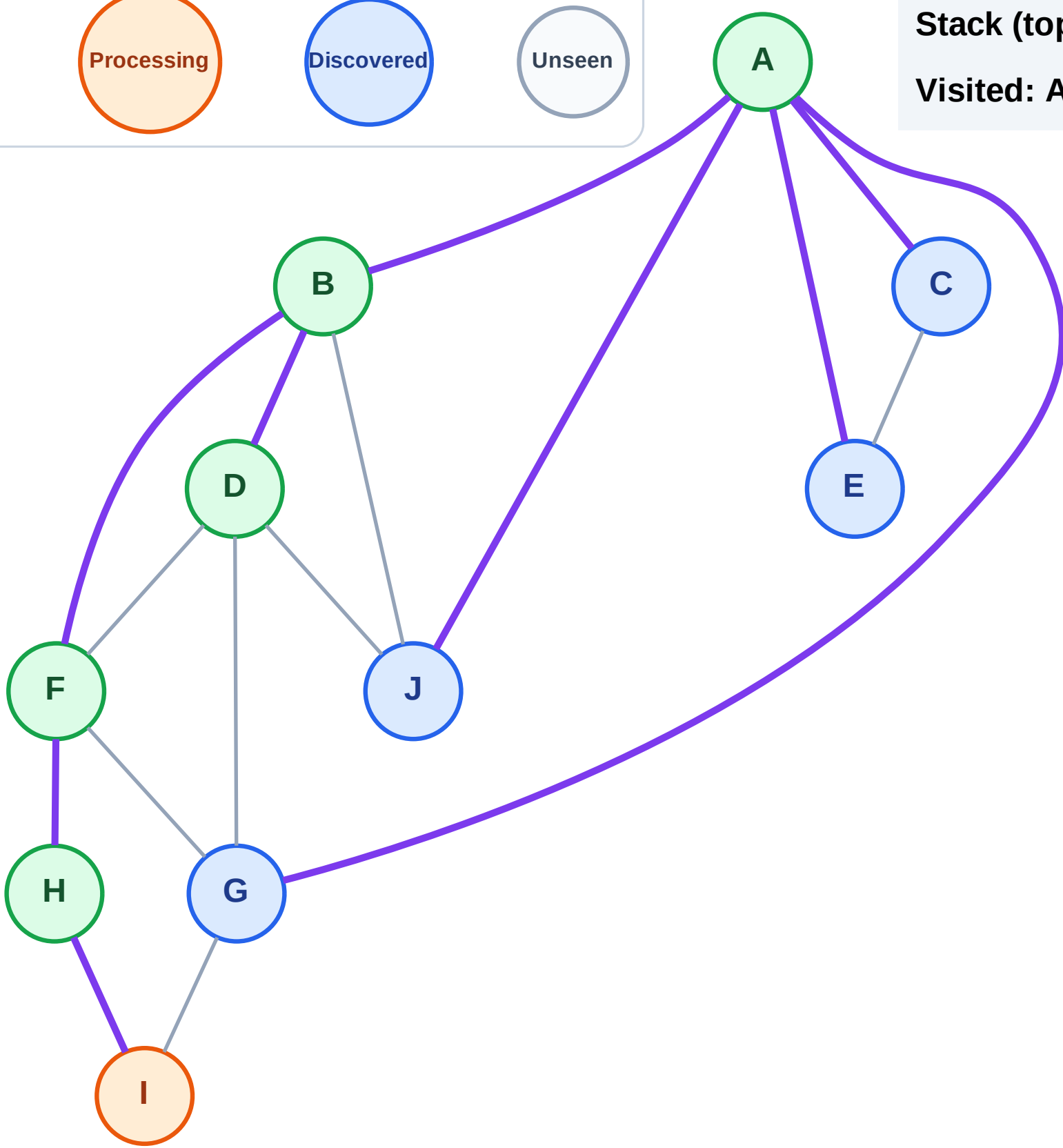
Step 12: Process node I

Legend



Stack (top at right): J | G | E | C

Visited: A, B, D, F, H



DFS Traversal

Step 13: No new neighbors from I

Legend

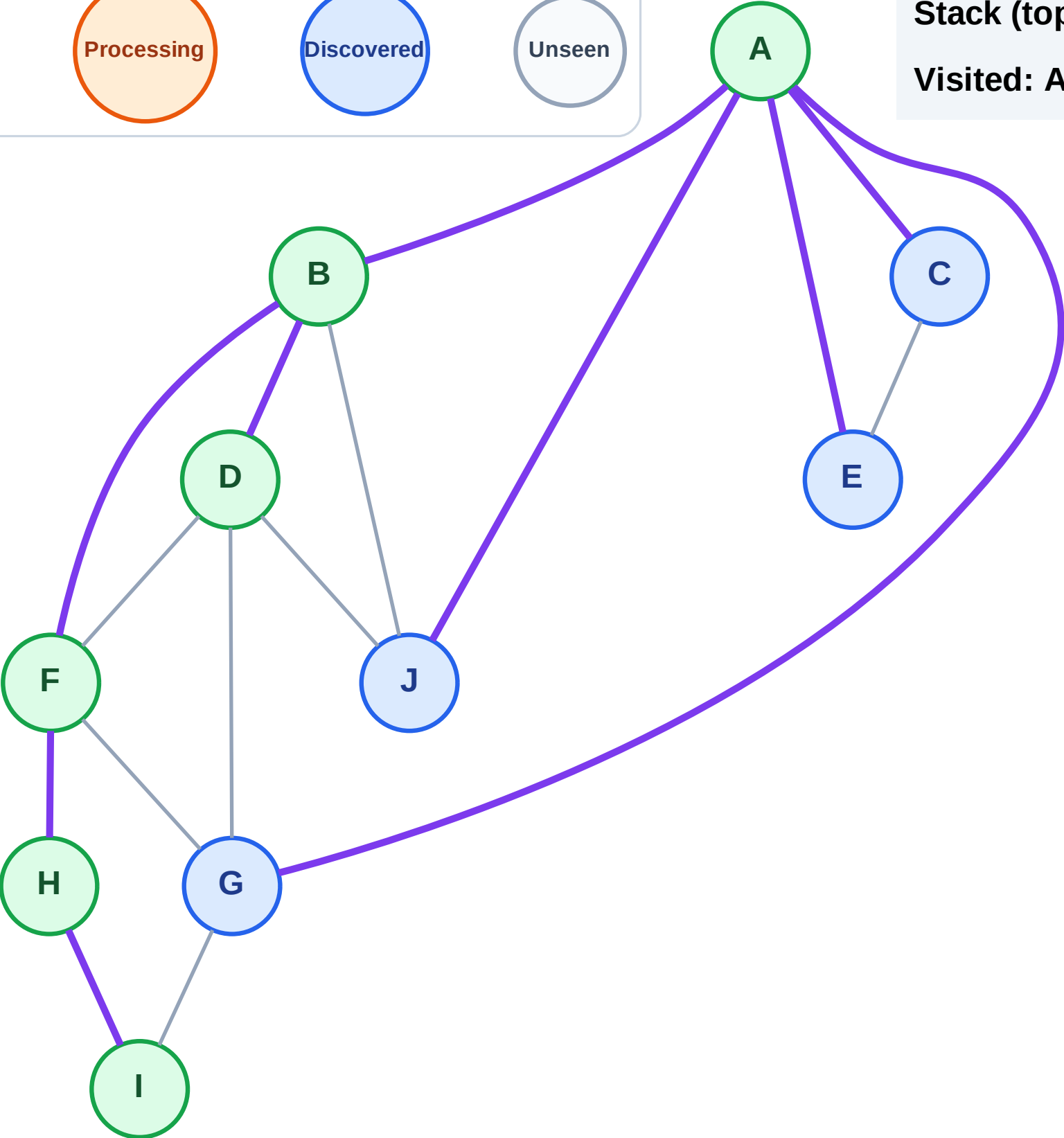
Complete

Processing

Discovered

Unseen

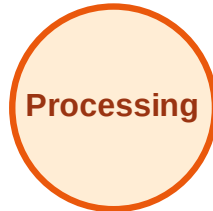
Stack (top at right): J | G | E | C
Visited: A, B, D, F, H, I



DFS Traversal

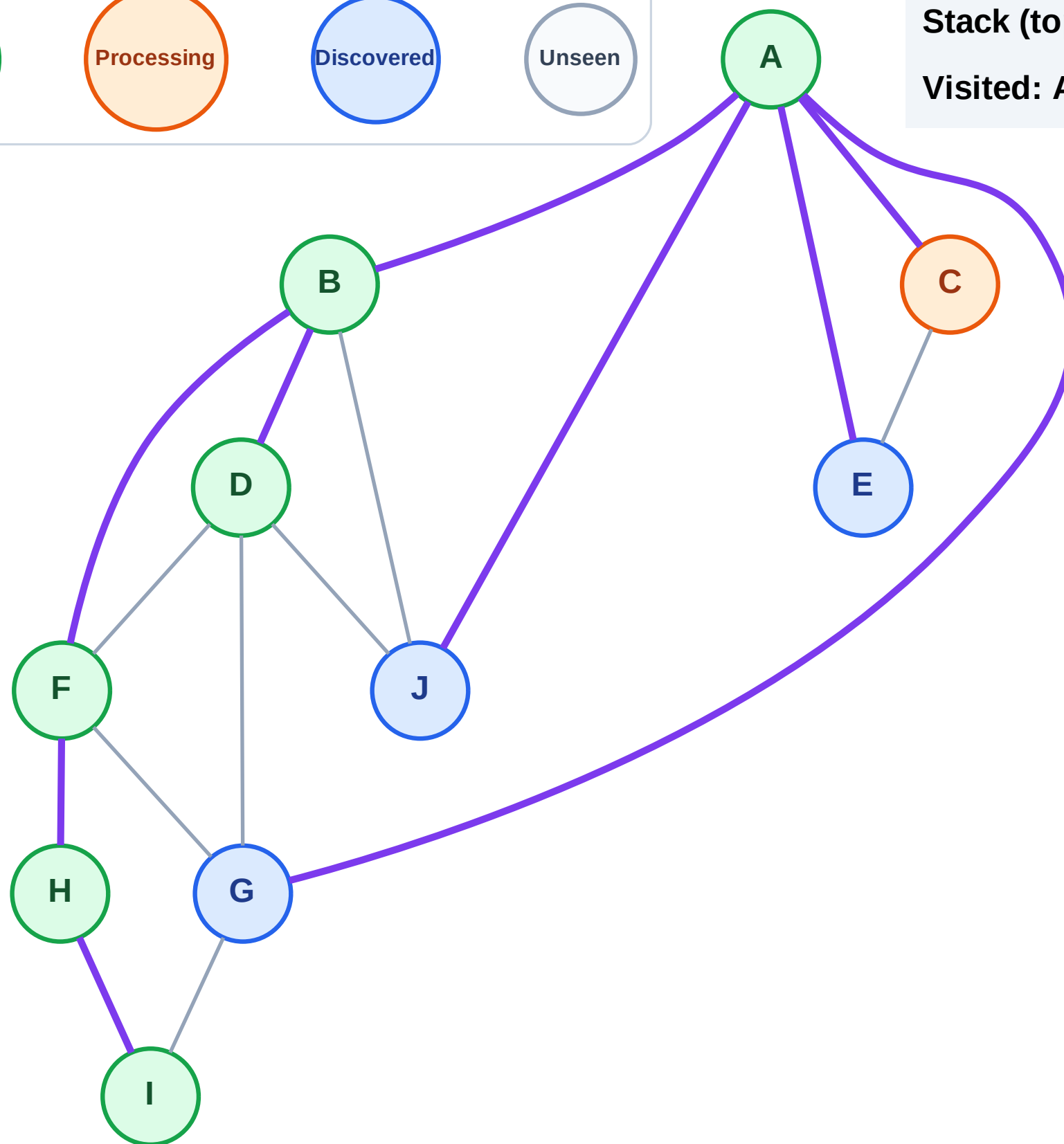
Step 14: Process node C

Legend



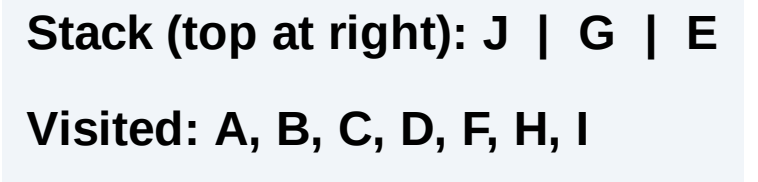
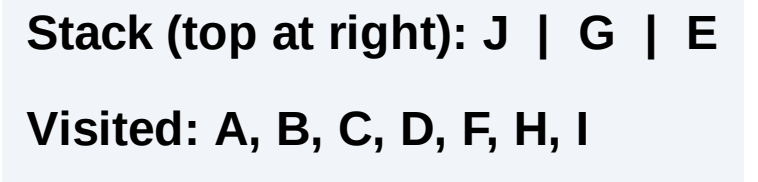
Stack (top at right): J | G | E

Visited: A, B, D, F, H, I



Step 15: No new neighbors from C

Legend



DFS Traversal

Step 16: Process node E

Legend

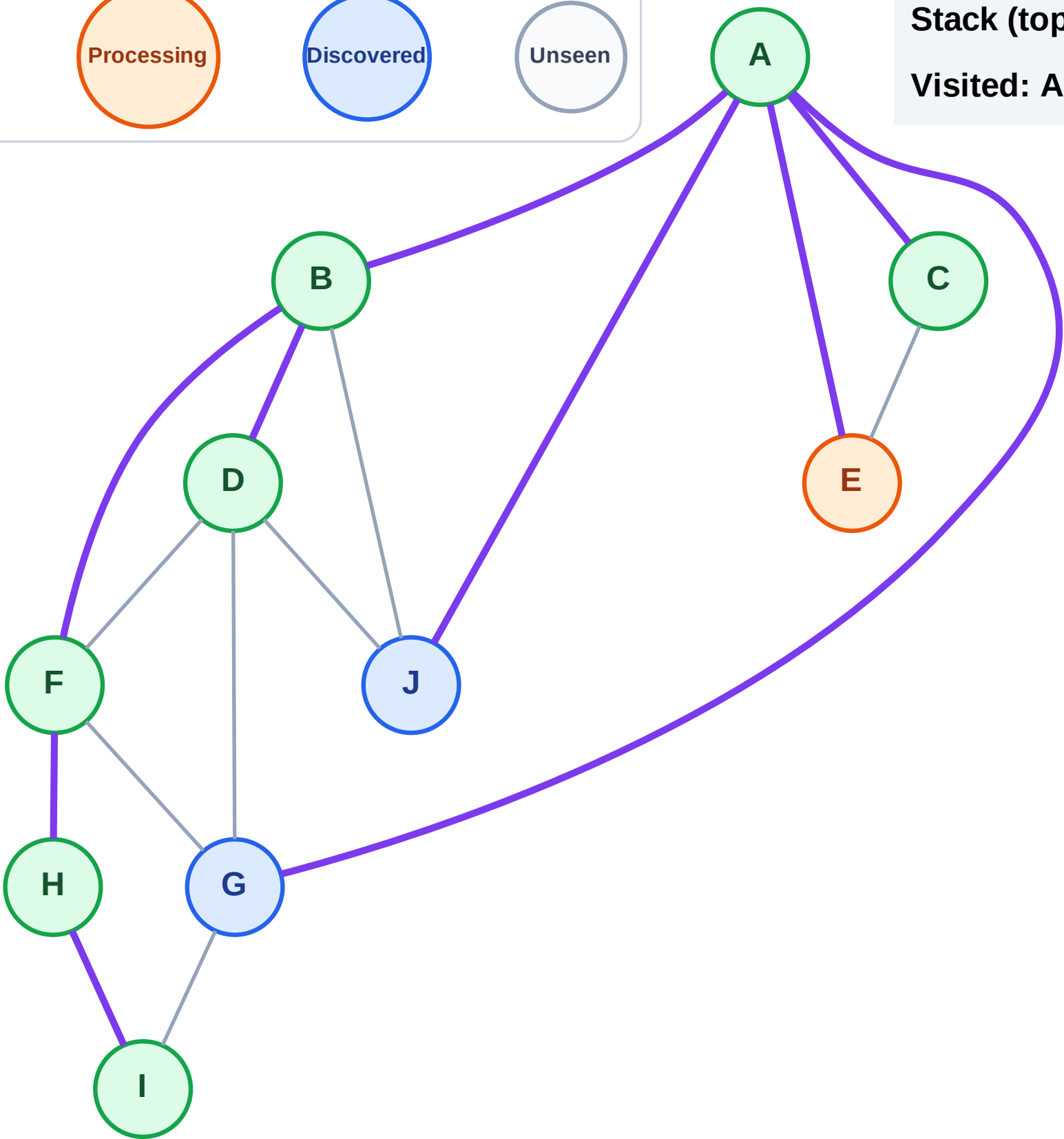
Complete

Processing

Discovered

Unseen

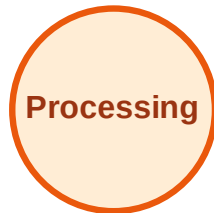
Stack (top at right): J | G
Visited: A, B, C, D, F, H, I



DFS Traversal

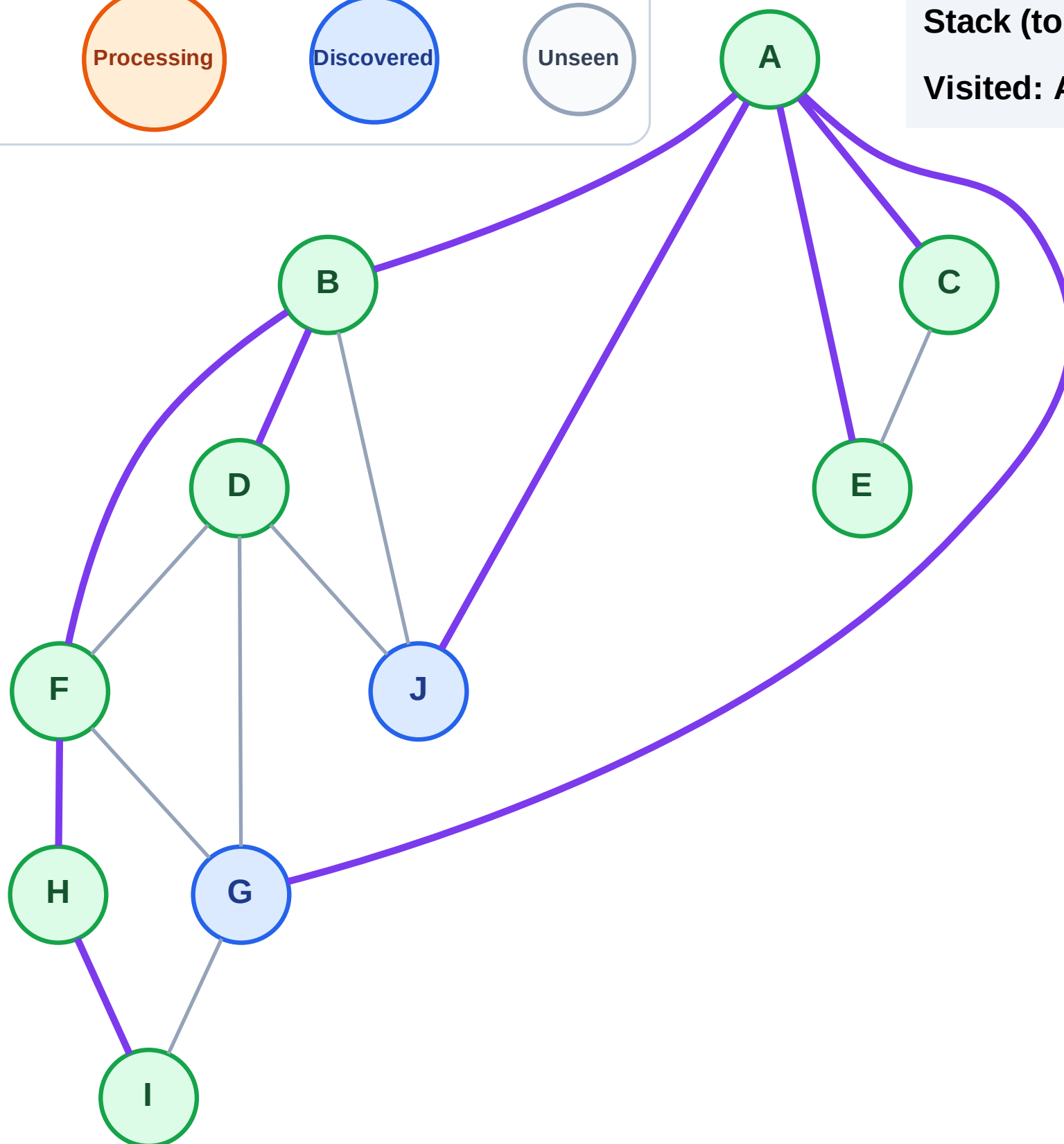
Step 17: No new neighbors from E

Legend



Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

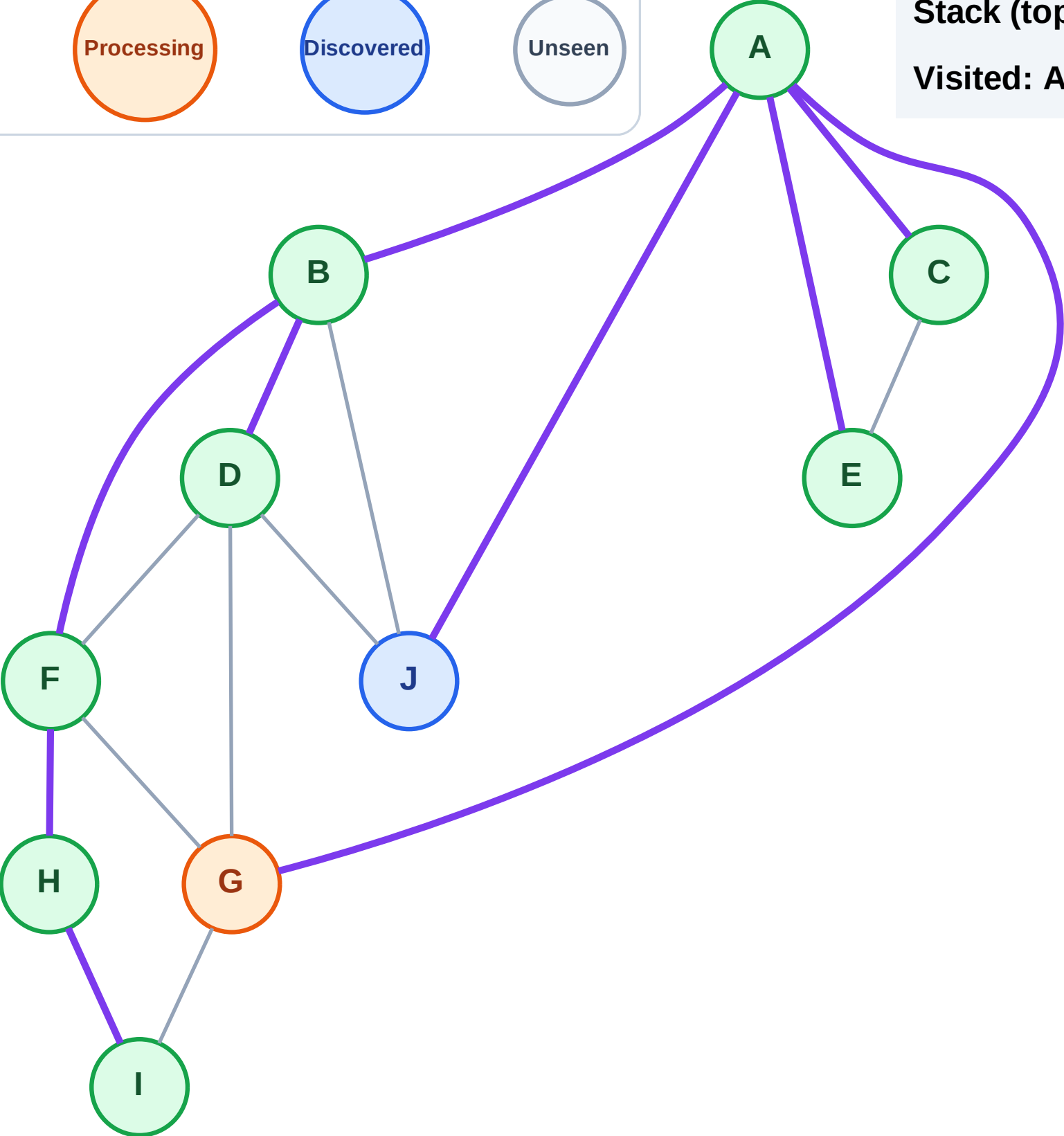
Complete

Processing

Discovered

Unseen

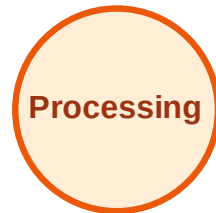
Stack (top at right): J
Visited: A, B, C, D, E, F, H, I



DFS Traversal

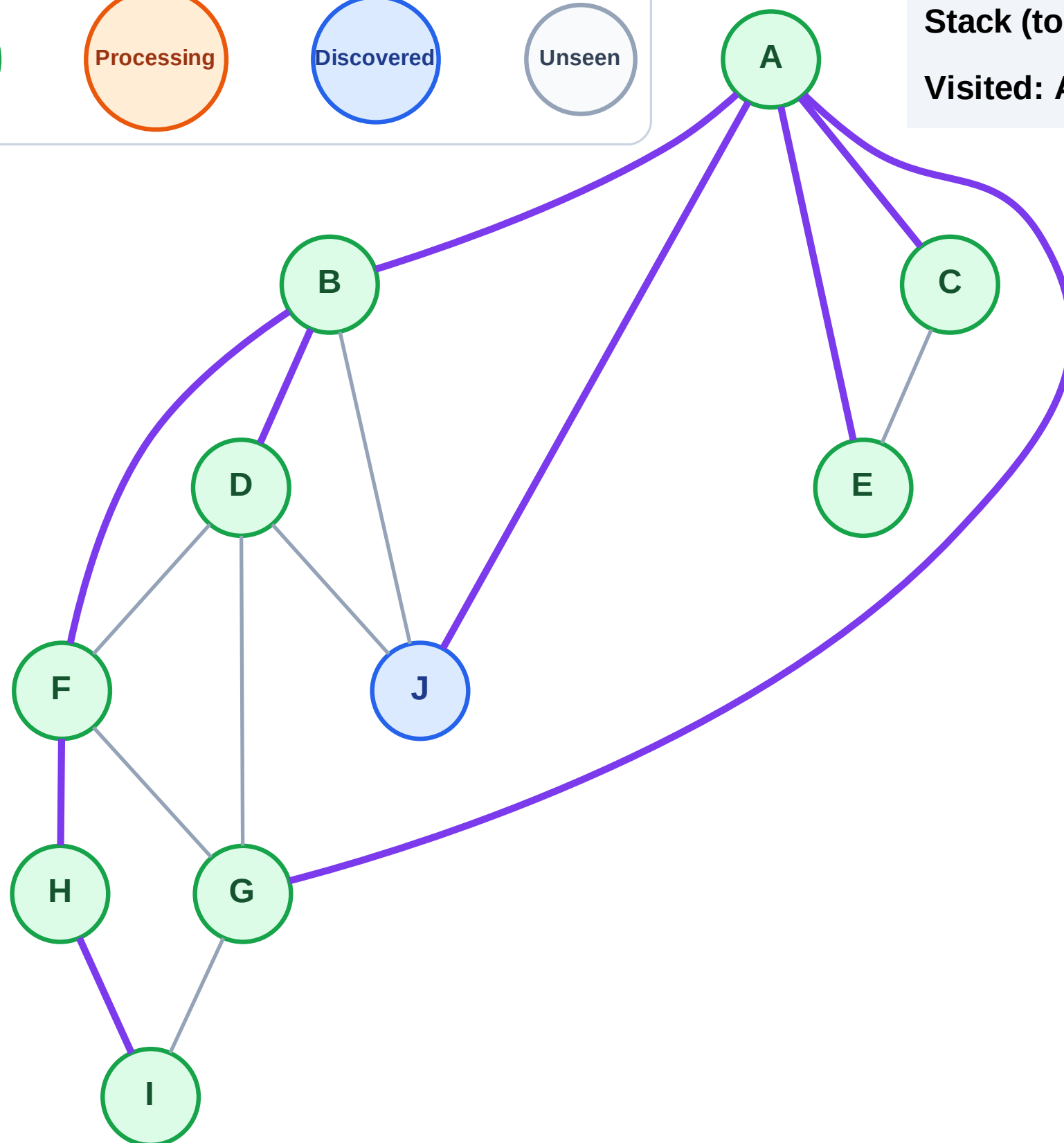
Step 19: No new neighbors from G

Legend



Stack (top at right): J

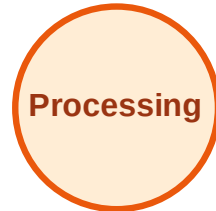
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

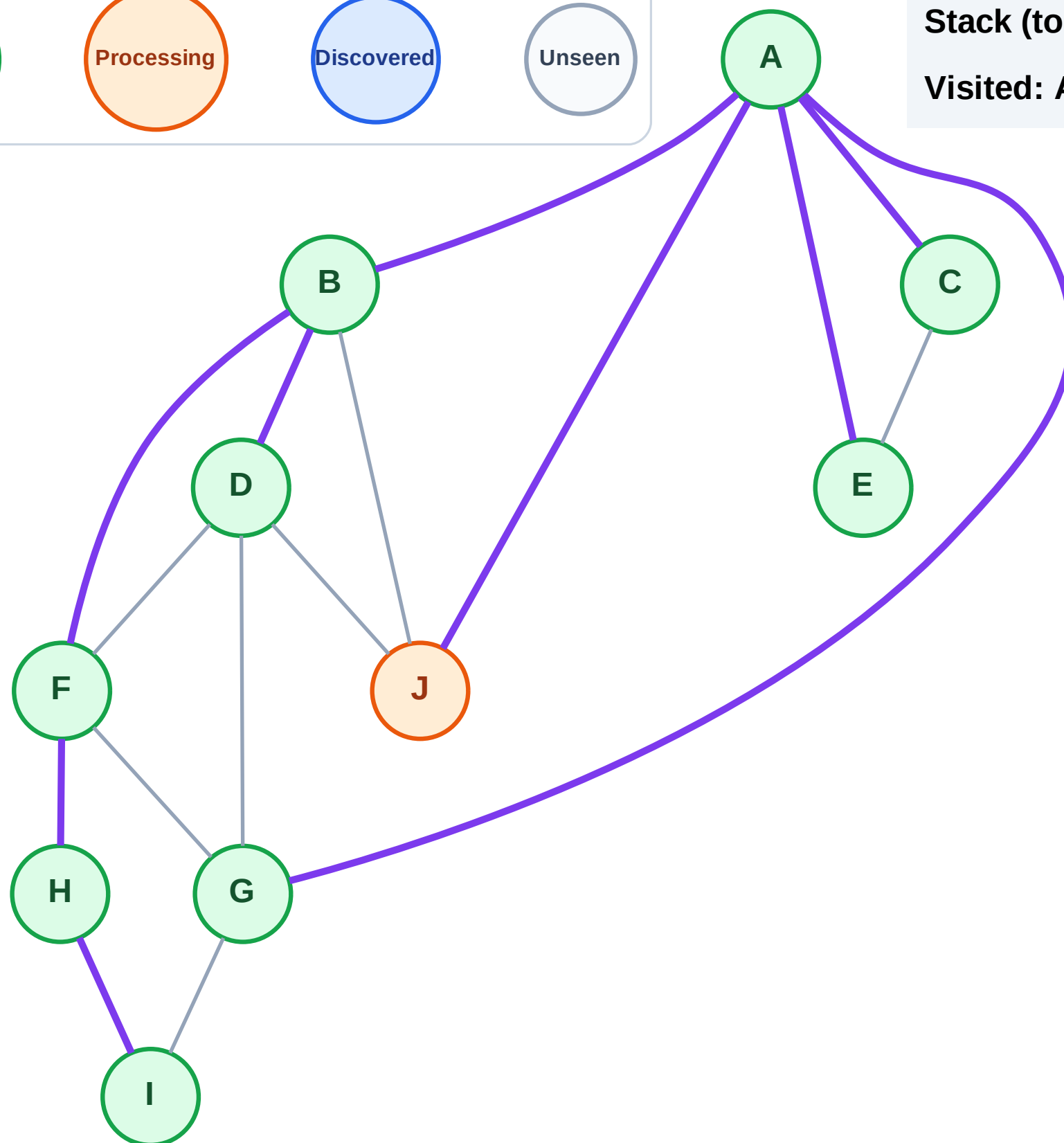
Step 20: Process node J

Legend



Stack (top at right): (empty)

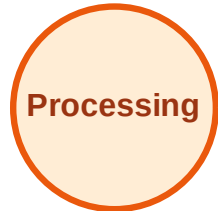
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

